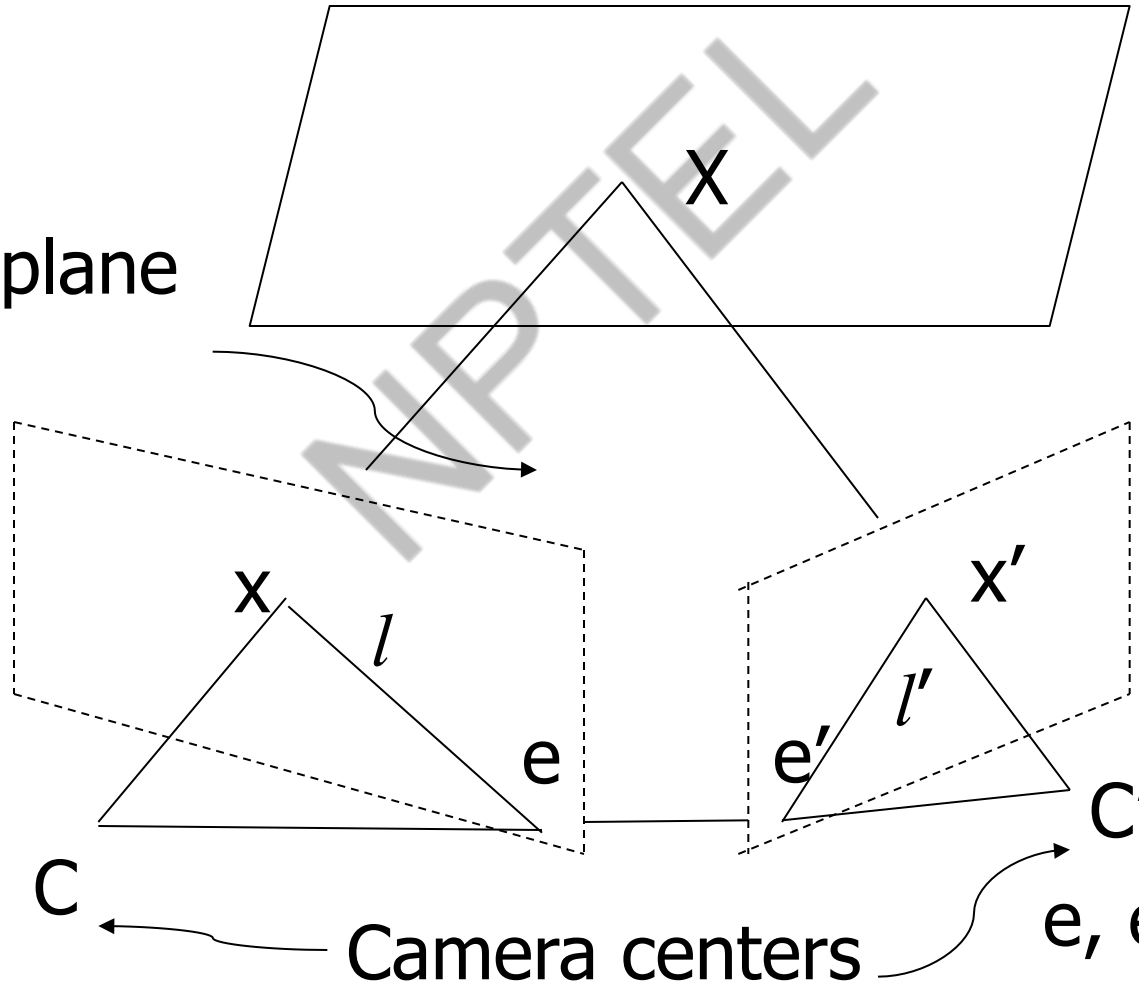
Stereo Geometry
(Week 04-05 Lectures: 16-23)

NPTEL

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Dept. of Computer Science and Engg.

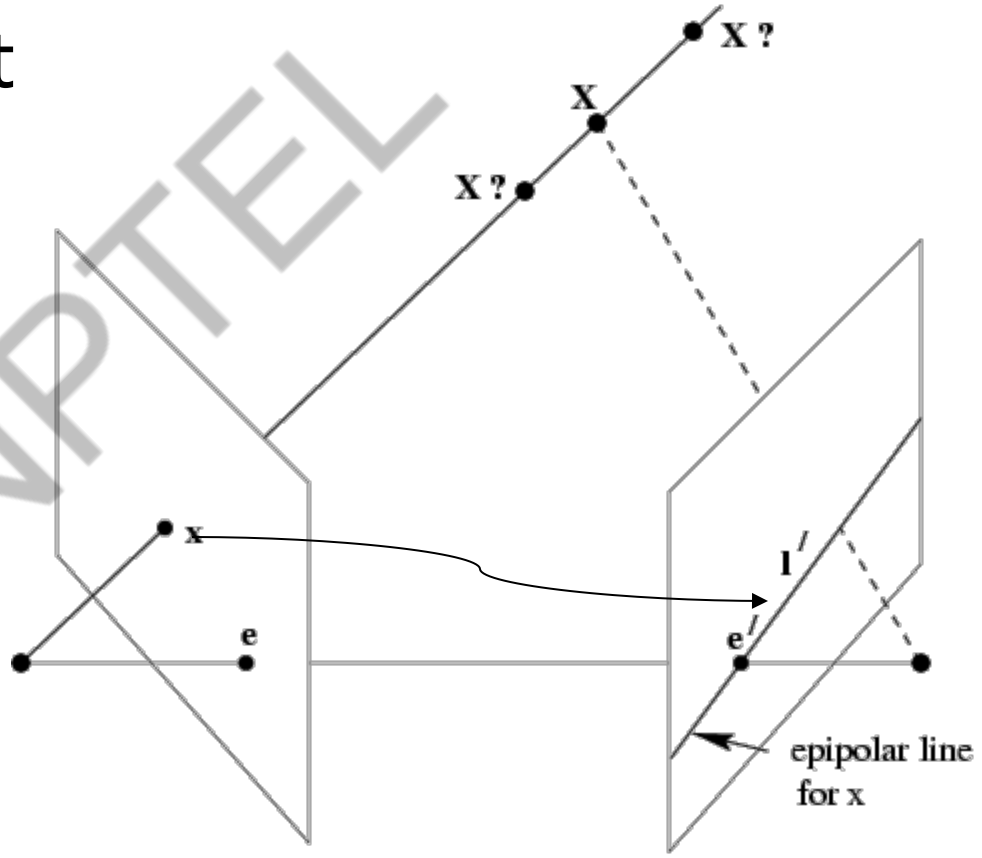
Stereo Set-up



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Epipolar geometry

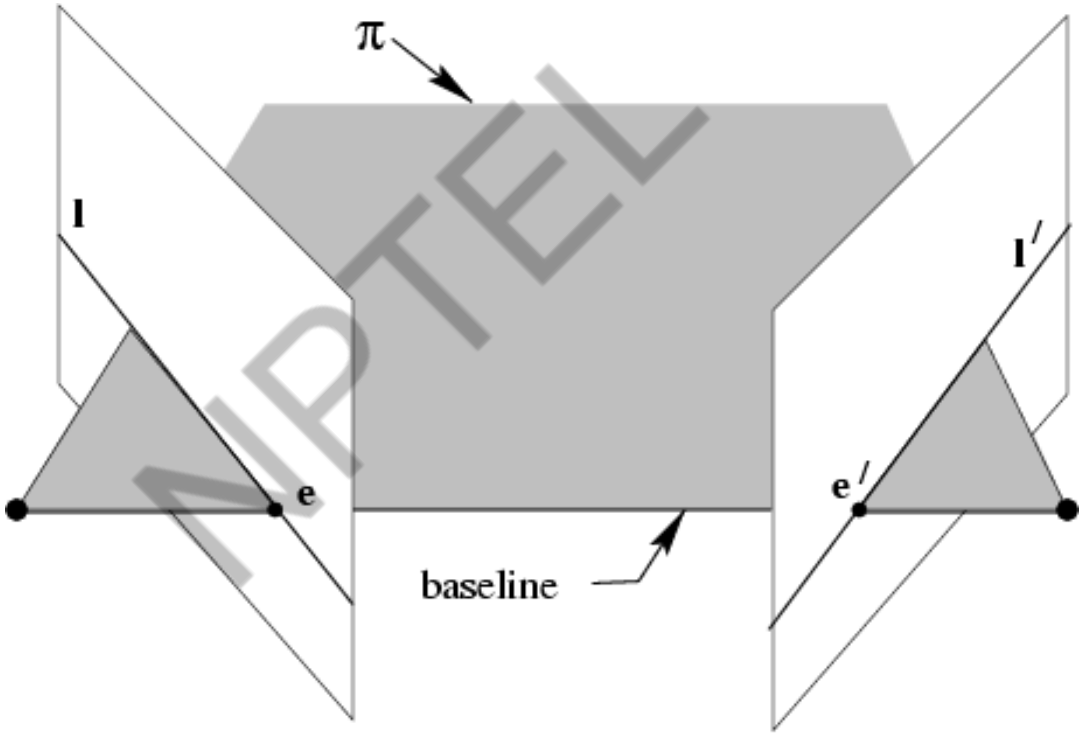
Corresponding point of x in the right image lies on l′.



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From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)

Epipolar geometry

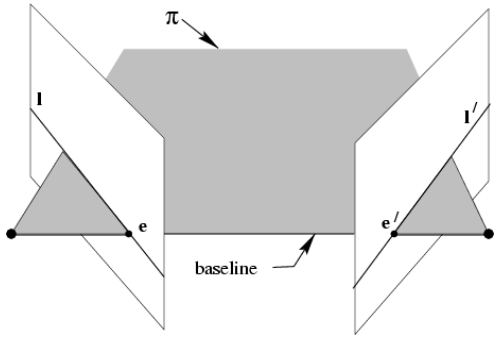


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All points on π project on l and l'

From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)

Epipolar geometry



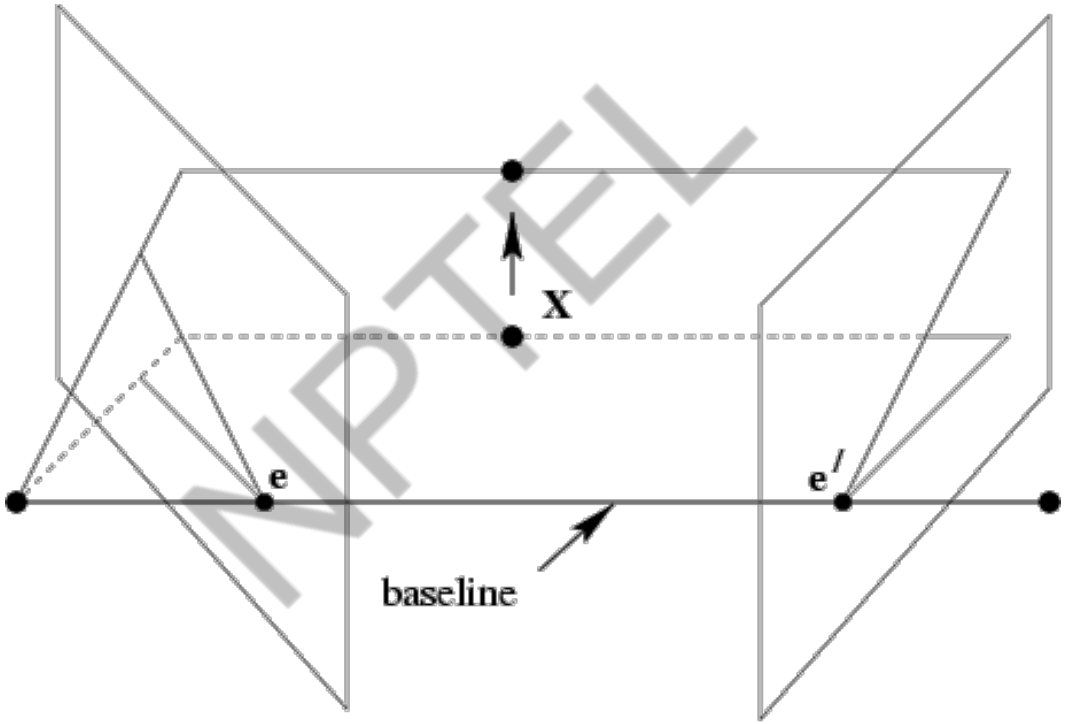
Epipoles e,e’
= intersection of baseline with image plane
= projection of projection center in other image
= vanishing point of camera motion direction

An epipolar plane: plane containing baseline (1-D family)

An epipolar line: intersection of epipolar plane with image (always come in corresponding pairs)

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Epipolar geometry



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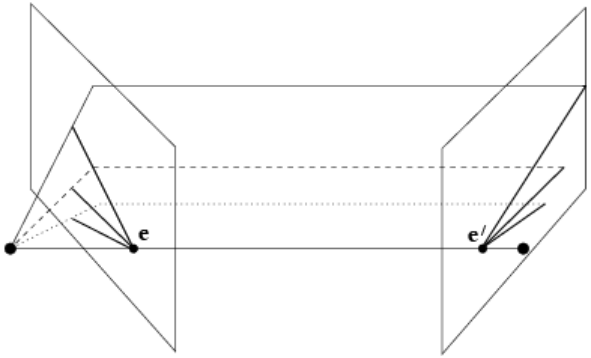
Family of planes π and lines l and l'
Intersection in e and e'

From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)

Example: converging cameras

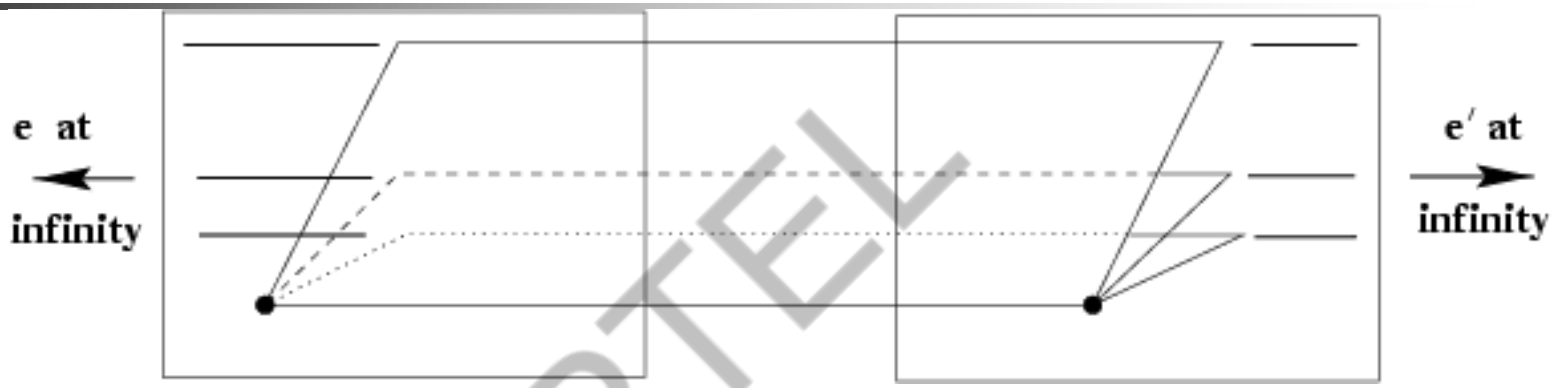






From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)

Example: motion parallel with image plane







From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)

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Epipolar Geometry

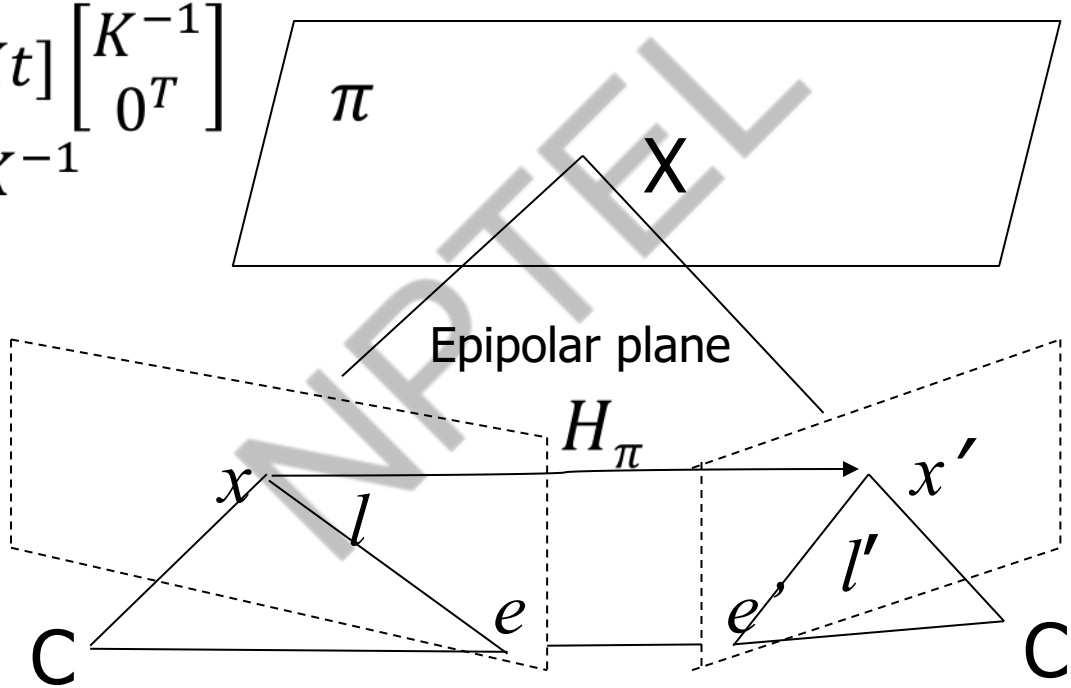
$$[e']_\times = \begin{bmatrix} 0 & -e_z & e_y \\ e_z & 0 & -e_x \\ -e_y & e_x & 0 \end{bmatrix}$$

$$H_\pi = [K'R|Kt]\begin{bmatrix} K^{-1} \\ 0^T \end{bmatrix} = K'RK^{-1}$$

$$x' = P'X = P'P^+x$$

$$P^+ = \begin{bmatrix} K^{-1} \\ 0^T \end{bmatrix}$$

For scene points lying on plane at infinity.



$$l' = e' \times x' = [e']_\times x' = [e']_\times H_\pi x = Fx$$

$P=K[I \mid 0]$

$P'=K'[R \mid t]$

$F=[e']_\times K'RK^{-1}$

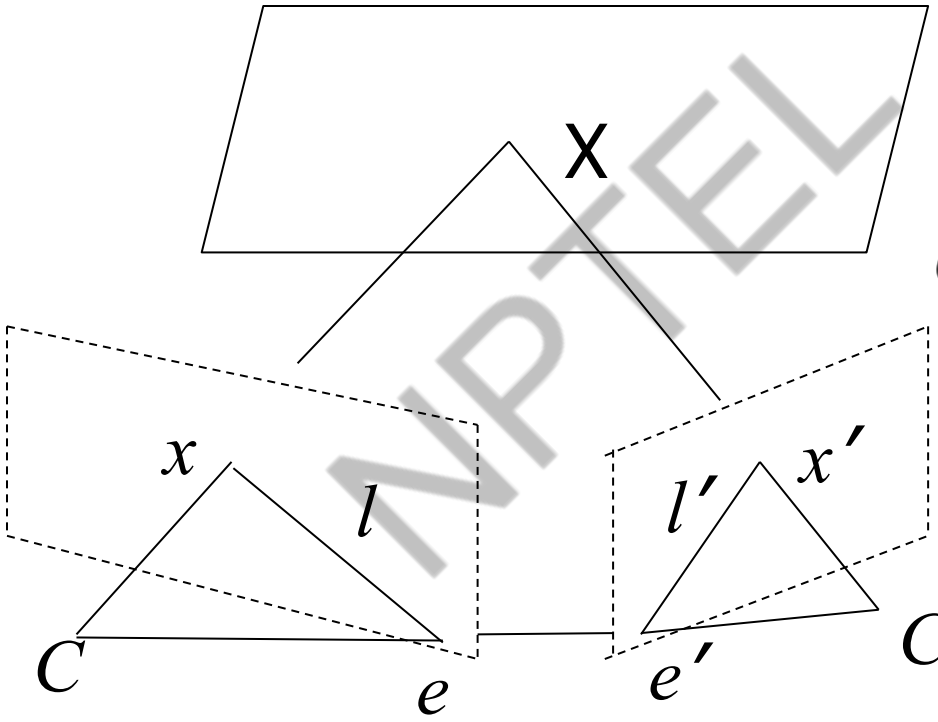
Coplanar: X, x, x', C, C′, e, e', l, l'

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Epipolar Geometry

Fundamental Matrix: F

$$l' = Fx$$



$$e = PC'$$

$$e' = P'C$$

$$e = -KR^T t \equiv KR^T t$$

$$e' = K't$$

$$l' = Fx$$

$$\Longrightarrow x'^T l' = 0$$

$$\Longrightarrow x'^T Fx = 0$$

$$\Longrightarrow x^T F^T x' = 0$$

$P = K[\, I \mid 0]$

$P' = K'[\, R \mid t]$

$$C = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$C' = \begin{bmatrix} -R^T t \\ 1 \end{bmatrix}$$

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Fundamental and Essential Matrices

$$P = K[I \mid 0]$$

$$P' = K'[R \mid t]$$

$$F = [e']_\times P' P^+$$

$$= [K't]_\times K' R K^{-1}$$

$$F = [m]_\times M$$

Say, $P=[I \mid 0]$, i.e. $K=I$

$$P' = K'[R \mid t] = [K'R \mid K't] = [M \mid m]$$

$$for\ K = I \quad and\ K' = I, \qquad F = [t]_\times R$$

Essential Matrix (E)

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Ex. 1

- Consider the following stereo imaging matrices given by $P = [I|0]$ (ref. camera) and P' as follows.

$$P' = \begin{bmatrix} 3 & 4 & 6 & 4 \\ 8 & 7 & 2 & 8 \\ 1 & 5 & 2 & 1 \end{bmatrix}$$

Compute the fundamental matrix of the system.

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Ans. 1

$$P' = \begin{bmatrix} 3 & 4 & 6 & 4 \\ 8 & 7 & 2 & 8 \\ 1 & 5 & 2 & 1 \end{bmatrix}$$

$$M = \begin{bmatrix} 3 & 4 & 6 \\ 8 & 7 & 2 \\ 1 & 5 & 2 \end{bmatrix} \qquad m = \begin{bmatrix} 4 \\ 8 \\ 1 \end{bmatrix} \qquad [m]_\times = \begin{bmatrix} 0 & -1 & 8 \\ 1 & 0 & -4 \\ -8 & 4 & 0 \end{bmatrix}$$

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$$F = [m]_\times M$$

$$= \begin{bmatrix} 0 & 33 & 14 \\ -1 & -16 & -2 \\ 8 & -4 & -40 \end{bmatrix}$$

Ex. 2

- Consider the following stereo imaging matrices given by P (ref. camera) and P′.

$$P = \begin{bmatrix} 3 & 2 & 4 & -2 \\ 8 & 6 & 0 & 4 \\ 9 & 5 & 7 & 3 \end{bmatrix} \qquad P' = \begin{bmatrix} 3 & 8 & 5 & 2 \\ 2 & 7 & 6 & -3 \\ 6 & 4 & 9 & 8 \end{bmatrix}$$

(a) Compute the fundamental matrix of the system.

(b) Given an image point (15,20) of the reference camera (P), compute the epipolar line and its two end image points of P′.

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Ans. 2 (a)

$$P = \underbrace{\begin{bmatrix} 3 & 2 & 4 \\ 8 & 6 & 0 \\ 9 & 5 & 7 \end{bmatrix}}_{M} \underbrace{\begin{bmatrix} -2 \\ 4 \\ 3 \end{bmatrix}}_{p_4} \qquad P' = \underbrace{\begin{bmatrix} 3 & 8 & 5 \\ 2 & 7 & 6 \\ 6 & 4 & 9 \end{bmatrix}}_{M'} \begin{bmatrix} 2 \\ -3 \\ 8 \end{bmatrix}$$

$$\tilde{C} = -M^{-1}p_4 \qquad M^{-1} = -\frac{1}{42}\begin{bmatrix} 42 & 6 & -24 \\ -56 & -15 & 32 \\ -14 & 3 & 2 \end{bmatrix}$$

$$C = \begin{bmatrix} \tilde{C} \\ 1 \end{bmatrix} = \frac{1}{42}\begin{bmatrix} -132 \\ 148 \\ 46 \\ 1 \end{bmatrix} \qquad e' = P'C = \begin{bmatrix} 26.23 \\ 21.94 \\ 13.09 \end{bmatrix} = 13.09 \begin{bmatrix} 2 \\ 1.67 \\ 1 \end{bmatrix}$$

$$H_\propto = M'M^{-1} \Rightarrow H_\propto = \begin{bmatrix} 9.33 & 2.07 & -4.62 \\ 9.33 & 1.78 & -4.48 \\ 2.33 & -0.07 & -0.05 \end{bmatrix}$$

$$F = [e']_\times H_\propto$$

$$[e']_\times = \begin{bmatrix} 0 & -1 & 1.67 \\ 1 & 0 & -2 \\ -1.67 & 2 & 0 \end{bmatrix}$$

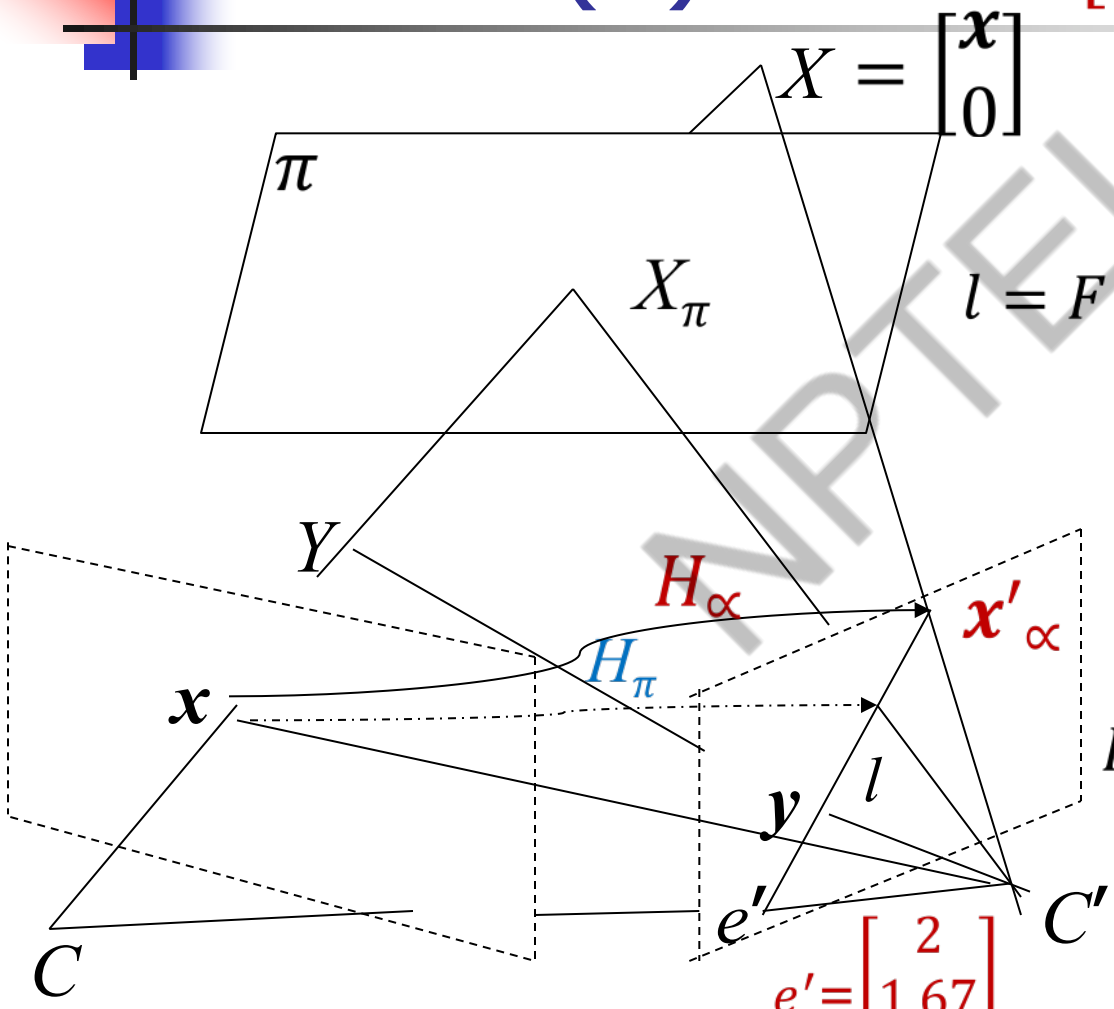
$$F = \begin{bmatrix} -5.41 & -1.91 & 4.40 \\ 4.67 & 2.21 & -4.52 \\ 2.99 & 0.09 & -1.19 \end{bmatrix}$$

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(b) Given an image point (15,20) of the reference camera (P), compute the epipolar line and its two end image points of P′.

Ans. 2(b)

$$F = \begin{bmatrix} -5.41 & -1.91 & 4.40 \\ 4.67 & 2.21 & -4.52 \\ 2.99 & 0.09 & -1.19 \end{bmatrix}$$



$$l = F\begin{bmatrix} 15 \\ 20 \\ 1 \end{bmatrix} = \begin{bmatrix} -114.92 \\ 109.76 \\ 45.44 \end{bmatrix} \equiv \begin{bmatrix} -2.53 \\ 2.42 \\ 1 \end{bmatrix}$$

$$x'_\infty = H_\infty \begin{bmatrix} 15 \\ 20 \\ 1 \end{bmatrix}$$

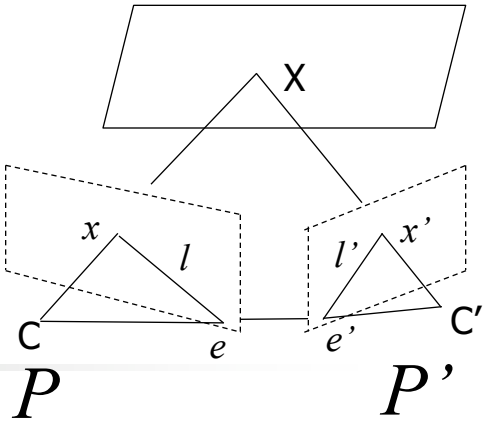
$$H_\infty = \begin{bmatrix} 9.33 & 2.07 & -4.62 \\ 9.33 & 1.78 & -4.48 \\ 2.33 & -0.07 & -0.05 \end{bmatrix}$$

$$e' = \begin{bmatrix} 2 \\ 1.67 \\ 1 \end{bmatrix}$$

$$x'_\infty = \begin{bmatrix} 176.81 \\ 171.24 \\ 33.52 \end{bmatrix} \equiv \begin{bmatrix} 5.27 \\ 5.11 \\ 1 \end{bmatrix}$$

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Fundamental Matrix: Properties



$$x'^T F x = x^T F^T x' = 0, \forall (x', x)$$

Transpose:

If F is fundamental matrix of (P,P'), F^T for (P',P).

Epipolar lines: For x, epipolar line $l'=Fx$.

For x', epipolar line $l=F^Tx'$.

F is a correlation.

→ Rank deficient.

→ Inverse does not exit.

Epipoles: $e'^T(Fx) = e^T(F^Tx') = (Fe)^Tx' = 0$

$e'^TF=0$ ➔ e′ is left NULL vector of F.

$Fe=0$ ➔ e is the right NULL vector of F.

Rank deficient:

det(F)=0 and F is a projective element → 7 d.o.f.

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Ex. 3

- Consider the following fundamental matrix.

$$F = \begin{bmatrix} 20 & 12 & 0 \\ 8 & -7 & 59 \\ -4 & 33 & -177 \end{bmatrix}$$

a) Given two points (5,8) and (7,-5) in the left image compute the corresponding epipolar lines in the right image.
b) Compute the right epipole.
c) Compute the left epipole.

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$$F = \begin{bmatrix} 20 & 12 & 0 \\ 8 & -7 & 59 \\ -4 & 33 & -177 \end{bmatrix}$$

Ans.3

- $l_1 = F.[\,5\ 6\ 1]^T = [172\quad 57\quad 1]^T$
- $l_2 = F.[\,7\ -5\ 1]^T = [80\quad 150\quad -370]^T$
- $e_R = l_1 \times l_2 = [-21240\quad 63720\quad 21240]^T$
  $= [-1\ \ 3\ \ 1]^T$

- e_L = Right zero of F

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$$F = \begin{bmatrix} 20 & 12 & 0 \\ 8 & -7 & 59 \\ -4 & 33 & -177 \end{bmatrix}$$

Ans.3 (contd.)

- e_L = Right zero of F

$$F = \begin{bmatrix} 20 & 12 & 0 \\ 8 & -7 & 59 \\ -4 & 33 & -177 \end{bmatrix}$$

$$\begin{bmatrix} 20 & 12 & 0 \\ 8 & -7 & 59 \end{bmatrix} \begin{bmatrix} e_{L1} \\ e_{L2} \\ 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} e_{L1} \\ e_{L2} \end{bmatrix} = -\begin{bmatrix} 20 & 12 \\ 8 & -7 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 59 \end{bmatrix} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}$$

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$$F = \begin{bmatrix} 20 & 12 & 0 \\ 8 & -7 & 59 \\ -4 & 33 & -177 \end{bmatrix}$$

Ans.3 (contd.)

- e_R= Right zero of F^T

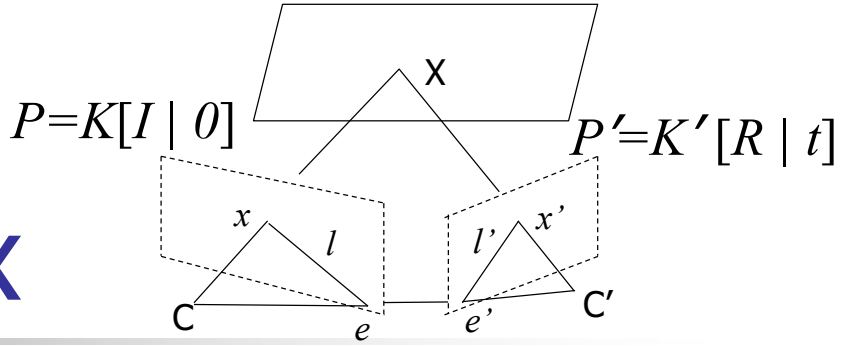
$$F^T = \begin{bmatrix} 20 & 8 & -4 \\ 12 & -7 & 33 \\ 0 & 59 & -177 \end{bmatrix}$$

$$\begin{bmatrix} 20 & 8 & -4 \\ 12 & -7 & 33 \end{bmatrix} \begin{bmatrix} e_{R1} \\ e_{R2} \\ 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} e_{R1} \\ e_{R2} \end{bmatrix} = -\begin{bmatrix} 20 & 8 \\ 12 & -7 \end{bmatrix}^{-1} \begin{bmatrix} -4 \\ 33 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

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Essential Matrix



Stereo geometry for calibrated cameras.

$x_c = K^{-1}x$ $\quad x'^T_c E x_c = 0$ $\quad x'_c = K'^{-1}x'$

Coordinates in calibrated image planes.

$x'^T F x = 0$

$\rightarrow (K'x'_c)^T F (Kx_c) = 0$

$\rightarrow x'^T_c K'^T F K x_c = 0$

$\rightarrow x'^T_c (K'^T F K) x_c = 0$

$E = K'^T F K$ $\quad E = [t]_x R$

$F = K'^{-T} E K^{-1}$

$= K'^{-T} [t]_x R K^{-1}$ 6 parameters

E

$Ee_c = e'_c E = 0$

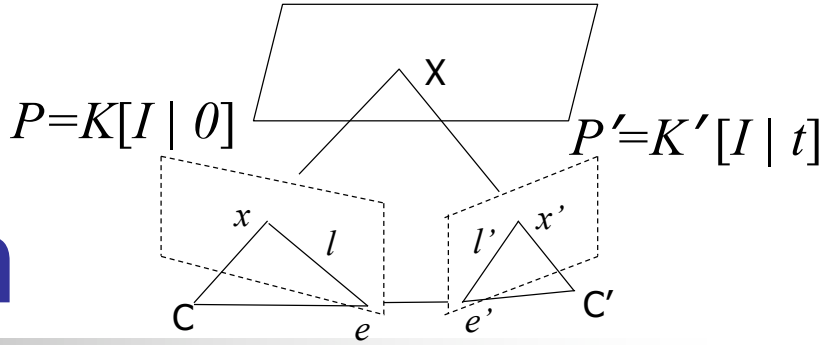
Rank: 2

$\det(E) = 0$

d.o.f.=5

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Pure translation



$F=[e']_x K'IK^{-1}$

$=[e']_x K'K^{-1}$

camera translation $\|^l$ to x-axis, $e'=[1\ 0\ 0]^T$ ➔ $F=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & -1\\ 0 & 1 & 0\end{bmatrix}$

For $K=K'$, $F=[e']_x$

$$[e']_\times=\begin{bmatrix}0 & -e_z & e_y\\ e_z & 0 & -e_x\\ -e_y & e_x & 0\end{bmatrix}$$

➔ $x'^T Fx=0$

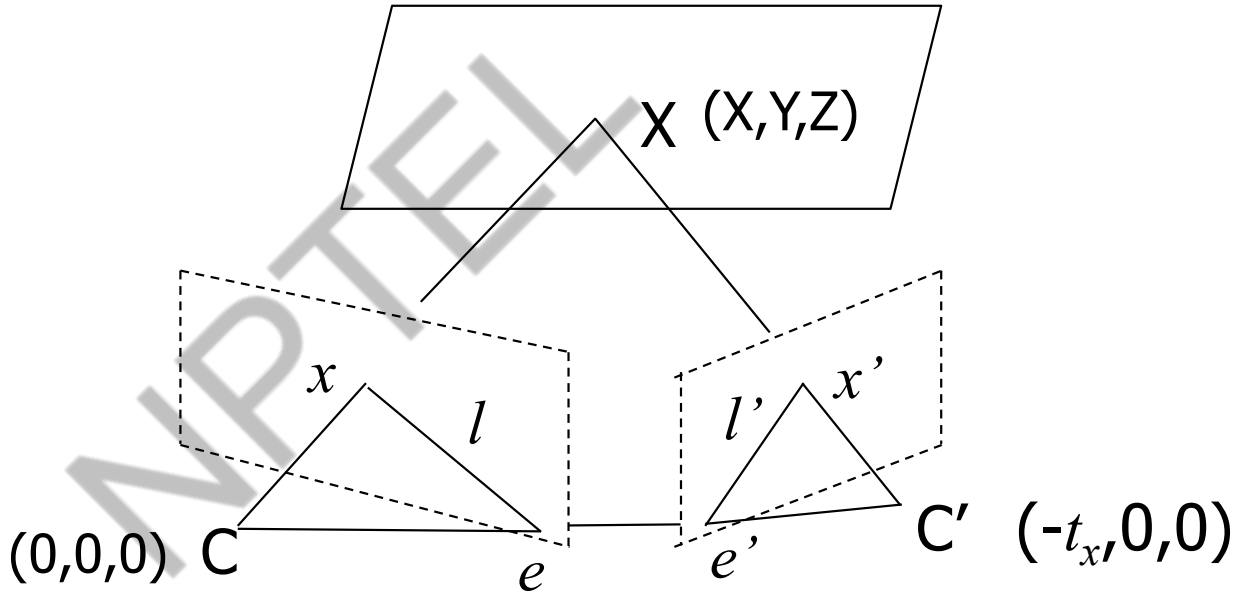
➔ $y'=y$

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Pure Translation: Computing Depth

$$\tilde{X} = ZK^{-1}x$$

$$\tilde{X} + t = ZK'^{-1}x'$$



For $K=K'$

$P=K[I \mid 0]$ $\quad P'=K'[I \mid t]$

$$ZK^{-1}(x' - x) = (\tilde{X} + t) - \tilde{X} = t \qquad t = \begin{bmatrix} t_x \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow x' = x + \frac{Kt}{Z} \quad \Rightarrow \quad Z = \frac{Kt}{x' - x}$$

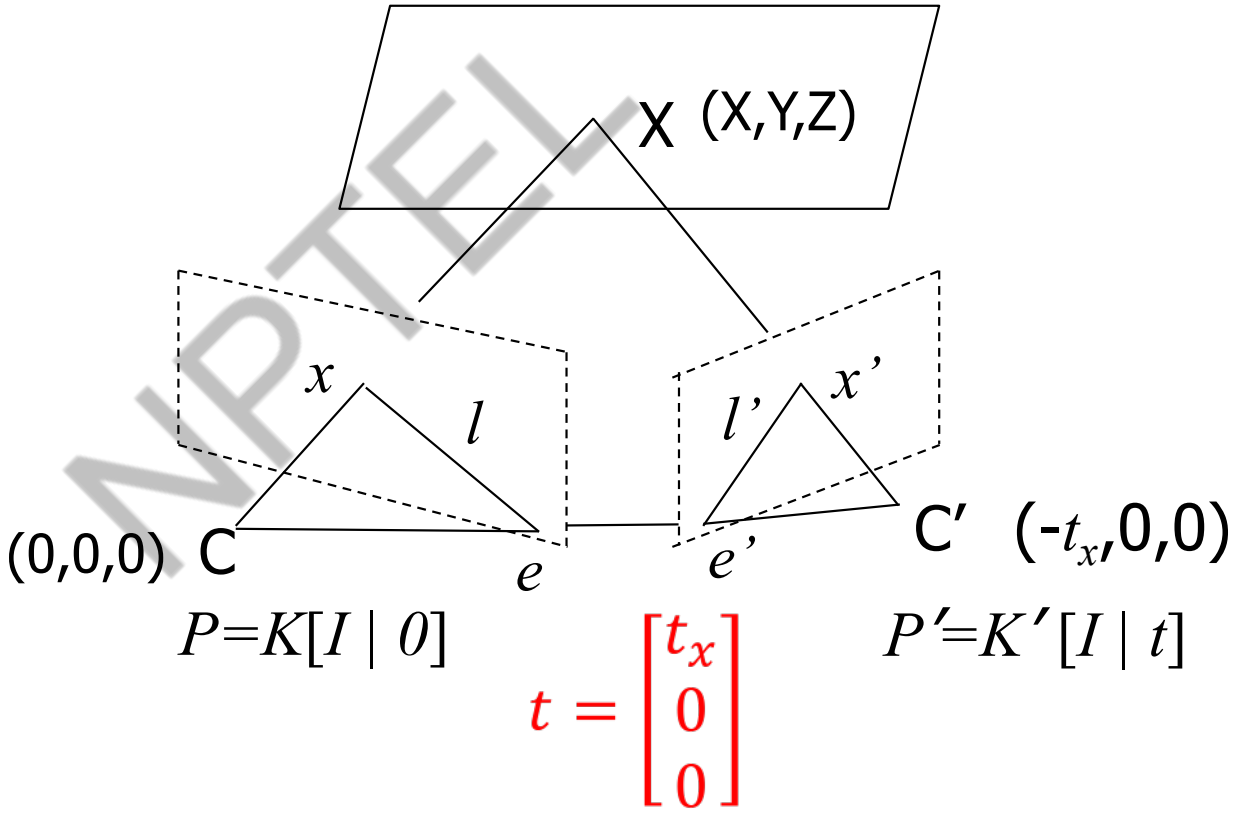
NPTEL

Pure Translation: Computing Depth

$$K = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$Z = \frac{Kt}{x' - x}$$

$$Z = \frac{f t_x}{x' - x}$$



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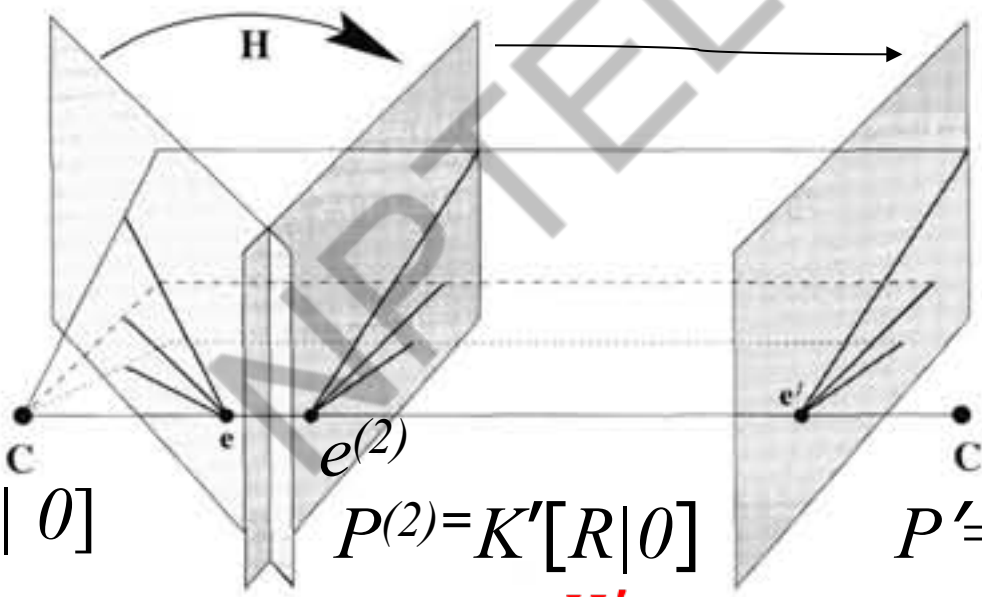
General Motion of Camera

$$x^{(2)} = K'[R|0]X$$
$$= K'R[I|0]X$$
$$= K'RK^{-1}K[I|0]X$$
$$= K'RK^{-1}x$$

Rotational Homography

$$H = K'RK^{-1}$$

$||^{l}$ stereo



$$x' = x^{(2)} + \frac{K't}{Z}$$
$$= K'RK^{-1}x + \frac{K't}{Z}$$

$P = K[I \mid 0]$

$P^{(2)} = K'[R|0]$

$P' = K'[R \mid t]$

$$Z = \frac{K't}{x' - K'RK^{-1}x}$$

From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)

General Motion of Camera

$$x^{(2)} = K[R|0]X$$
$$= KR[I|0]X$$
$$= KRK^{-1}K[I|0]X$$
$$= KRK^{-1}x$$

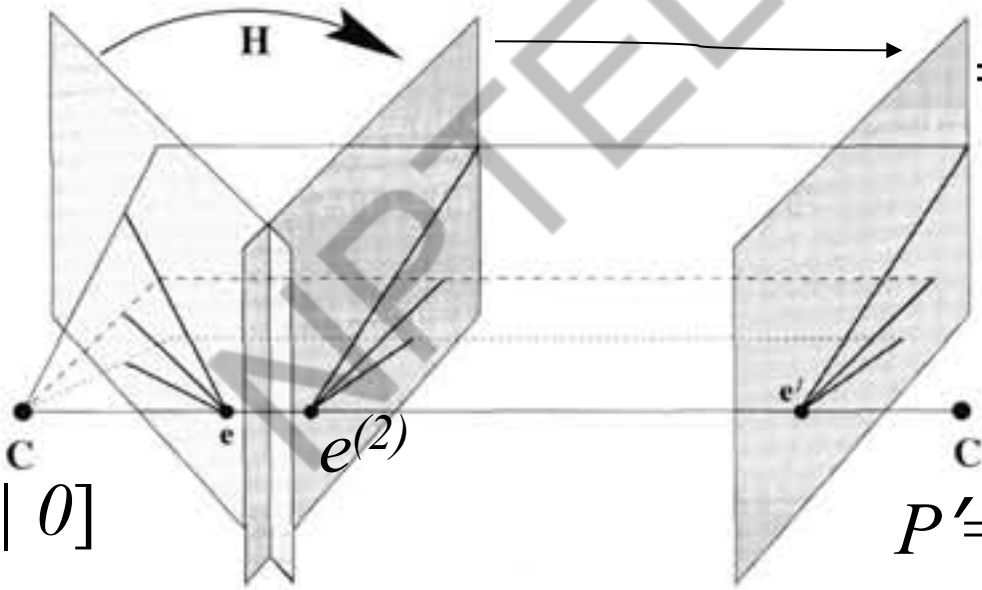
Rotational Homography

$H = KRK^{-1}$

$||^{l}$ stereo

$$x' = x^{(2)} + \frac{K't}{Z}$$

$$= KRK^{-1}x + \frac{K't}{Z}$$



$P = K[I \mid 0]$

$P' = K'[R \mid t]$

$$Z = \frac{K't}{x' - KRK^{-1}x}$$

From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)

Ex. 4

- Consider a stereo set-up with P and P′ (camera matrices for left and right camera) as given below.

$$P = \begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \qquad P' = \begin{bmatrix} 6 & 0 & 0 & 10 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

If the image coordinates of a 3-D point are (6,8) and (9.33,8) in left and right cameras, compute its depth (z-coordinate) in the 3D.

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Ans. 4

- P=K[I|0] and P′=K[I|t]

$$K = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad t = \begin{bmatrix} 10/6 \\ 0 \\ 0 \end{bmatrix}$$

- **x**′=**x**+(**Kt**)/Z
- Z=(6x10/6)/(x′-x)

  =10/(9.33-6)=3

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Estimation of Fundamental Matrix

$$F = \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix}$$

$$\mathrm{x}'^{\mathrm{T}} \mathrm{F}\mathrm{x} = 0$$

$$x'xf_{11} + x'yf_{12} + x'f_{13} + y'xf_{21} + y'yf_{22} + y'f_{23} + xf_{31} + yf_{32} + f_{33} = 0$$

$$[x'x, x'y, x', y'x, y'y, y', x, y, 1][f_{11}, f_{12}, f_{13}, f_{21}, f_{22}, f_{23}, f_{31}, f_{32}, f_{33}]^{\mathrm{T}} = 0$$

$$\begin{bmatrix} x'_1 x_1 & x'_1 y_1 & x'_1 & y'_1 x_1 & y'_1 y_1 & y'_1 & x_1 & y_1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x'_n x_n & x'_n y_n & x'_n & y'_n x_n & y'_n y_n & y'_n & x_n & y_n & 1 \end{bmatrix} \mathrm{f} = 0$$

$$\mathrm{Af} = 0$$

- Solution up to scale.
- Minimum 8 point correspondences.
- Use of DLT (for 7 point correspondences from linear combination of smallest and second smallest eigen vectors.

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Estimation of Fundamental Matrix

$$\begin{bmatrix} x_1x_1' & y_1x_1' & x_1' & x_1y_1' & y_1y_1' & y_1' & x_1 & y_1 & 1 \\ x_2x_2' & y_2x_2' & x_2' & x_2y_2' & y_2y_2' & y_2' & x_2 & y_2 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_nx_n' & y_nx_n' & x_n' & x_ny_n' & y_ny_n' & y_n' & x_n & y_n & 1 \end{bmatrix} \begin{bmatrix} f_{11} \\ f_{12} \\ f_{13} \\ f_{21} \\ f_{22} \\ f_{23} \\ f_{31} \\ f_{32} \\ f_{33} \end{bmatrix} = 0$$

~10000 ~10000 ~100 ~10000 ~10000 ~100 ~100 ~100 1

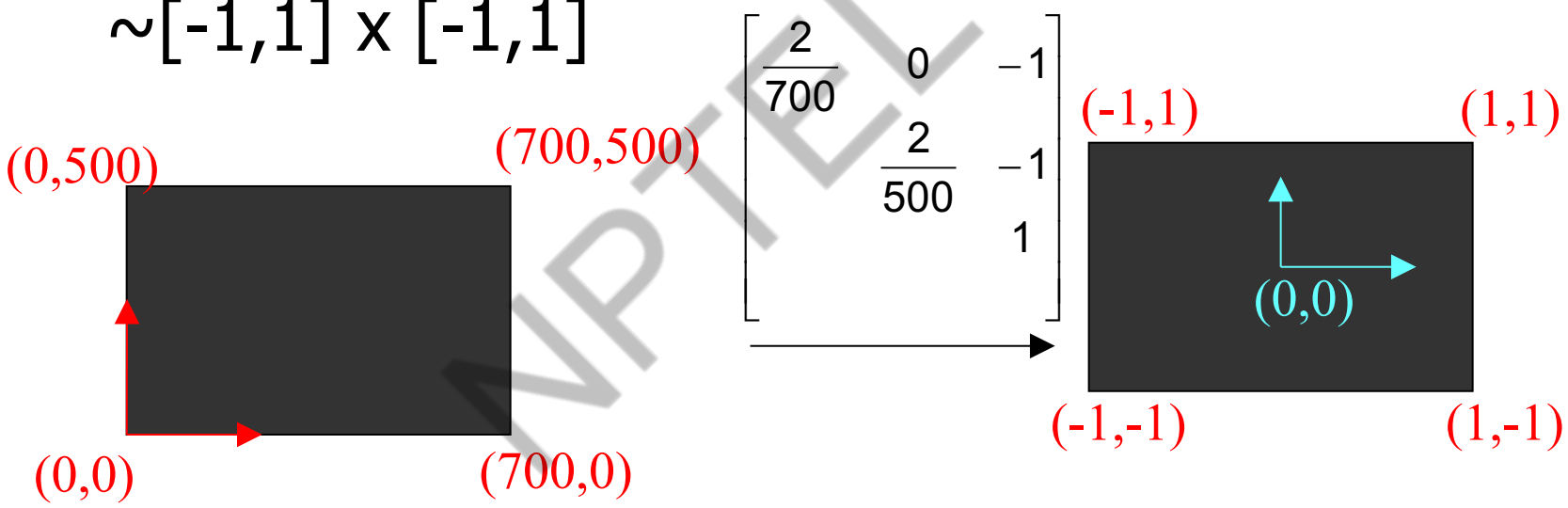


Orders of magnitude difference
Between column of data matrix
→ least-squares yield poor results

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The normalized 8-point algorithm

Transform image to
~[-1,1] x [-1,1]



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Least squares yields good results
(Hartley, PAMI´97)

The singularity constraint

$$\det F = 0 \quad \text{rank } F = 2$$

SVD from linearly computed F matrix (rank 3)

$$F = U\begin{bmatrix}\sigma_1 & & \\ & \sigma_2 & \\ & & \sigma_3\end{bmatrix}V^T = U_1\sigma_1 V_1^T + U_2\sigma_2 V_2^T + U_3\sigma_3 V_3^T$$

$$\min \|F - F'\|_F$$

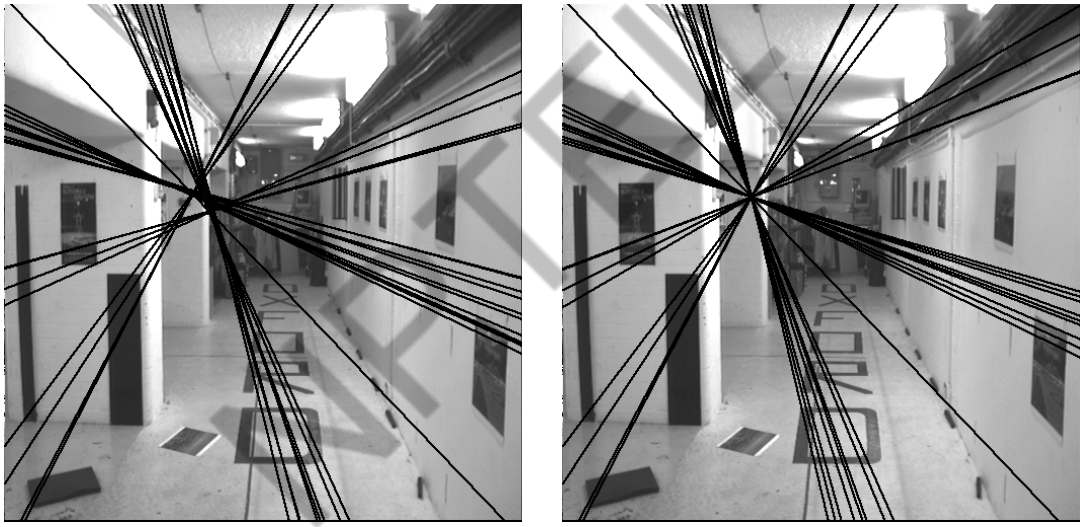
Compute closest rank-2 approximation

$$F' = U\begin{bmatrix}\sigma_1 & & \\ & \sigma_2 & \\ & & 0\end{bmatrix}V^T = U_1\sigma_1 V_1^T + U_2\sigma_2 V_2^T$$

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The singularity constraint

Nonsingular F



Singular F

Non-singular F causes epipolar lines not converging.

From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)

The singularity constraint for Essential Matrix

$\det(E)=0$

Estimate $\hat{E}$ by any of the techniques used for F.

Perform SVD of $\hat{E}$.

$\hat{E}=UDV^T$ Where $D=\text{diag}(a,b,c)$ $a \geq b \geq c$

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For essential matrix, two singular values are the same.

$$\Rightarrow \hat{E} = U\hat{D}V^T \qquad where\ \hat{D} = \left(\frac{a+b}{2}, \frac{a+b}{2}, 0\right)$$

The minimum case – 7 point correspondences

$$\begin{bmatrix} x'_1 x_1 & x'_1 y_1 & x'_1 & y'_1 x_1 & y'_1 y_1 & y'_1 & x_1 & y_1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x'_7 x_7 & x'_7 y_7 & x'_7 & y'_7 x_7 & y'_7 y_7 & y'_7 & x_7 & y_7 & 1 \end{bmatrix} \mathrm{f} = 0$$

$$A = U_{7x7}\,\mathrm{diag}(\sigma_1,\ldots,\sigma_7,0,0)\,V_{9x9}{}^{T}$$

Last two column vectors of V also provide F1 and F2.

F_1, F_2 → Eigen vectors corresponding to two zero's. The solution is $F_1+\lambda F_2$.

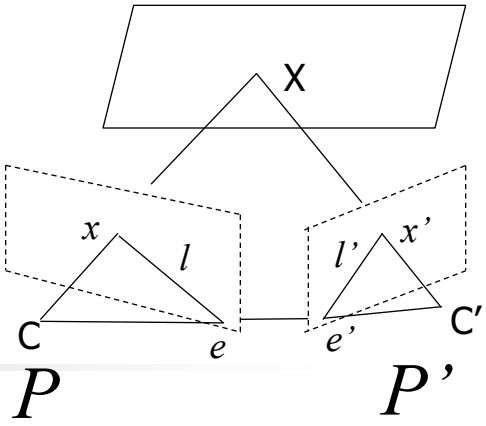
But $F_1+\lambda F_2$ not automatically rank 2.

Solve for λ from $\det(F_1+\lambda F_2) = 0$.

As it is a cubic polynomial, there are 1 or 3 solutions.

NPTEL

Parametric representation of F



Over parameterization: $F=[t]_x M$ ➔ $\{t,M\}$ → 12 params.

Epipolar parameterization:

$$F = \begin{bmatrix} a & b & \alpha a + \beta b \\ c & d & \alpha c + \beta d \\ e & f & \alpha e + \beta f \end{bmatrix} \qquad \{a, b, c, d, e, f, \alpha, \beta\}$$

Left epipole $e = [\alpha \quad \beta \quad -1]^T$

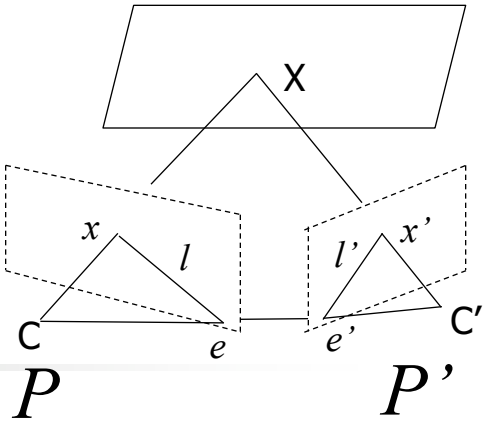
Both epipoles as parameters $\{a, b, c, d, \alpha, \beta, \alpha', \beta'\}$

$$F = \begin{bmatrix} a & b & \alpha a + \beta b \\ c & d & \alpha c + \beta d \\ \alpha' a + \beta' c & \alpha' b + \beta' d & \alpha\alpha' a + \alpha\beta' c + \alpha'\beta b + \beta\beta' d \end{bmatrix}$$

Epipoles $e = [\alpha \quad \beta \quad -1]^T$ $e' = [\alpha' \quad \beta' \quad -1]^T$

NPTEL

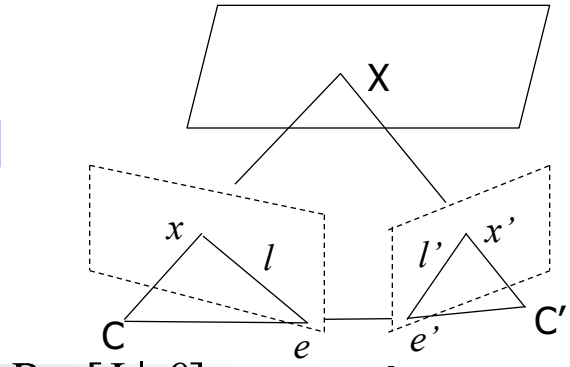
Retrieving the camera matrices from F



- F only depends on projective properties of P and P'.
- Independent of choice of world frame.
- (P,P') → F (unique)
- F→ (P,P') (?)
- Given a homography H (4x4 non-singular matrix) in P^3, if (P,P') → F, then $(PH,P'H)$→ F.
  Proof: $PX \leftrightarrow P'X$
  → $(PH)(H^{-1}X) \leftrightarrow (P'H)(H^{-1}X)$
- *F does not uniquely map to (P,P').*

NPTEL

Retrieving the camera matrices from F



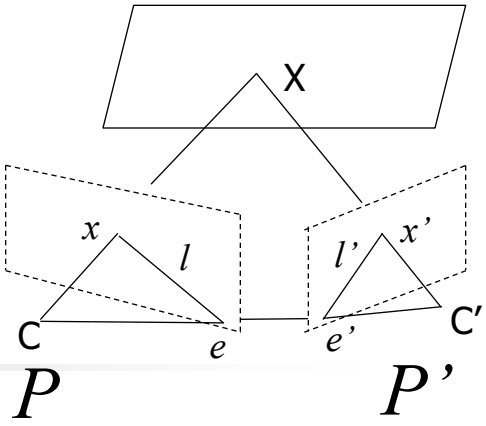
$P=[I \mid 0]$ $P'=[M \mid m]$

F=?

- $P=[I \mid 0]$ & $P'=[M \mid m]$ → $F=[m]_x M$.
- If F derived from both (P_1,P_1') and (P_2,P_2'), there exists 4x4 H s.t. $P_2=P_1H$ & $P_2'=P_1'H$.
- d.o.f. of P + d.o.f. of P'=22
- d.o.f. of H = 15
- d.o.f. of F=22-15=7

NPTEL

Retrieving the camera matrices from F



- F corresponds to (P,P'), iff $P'^{T}FP$ is *skew symmetric*.

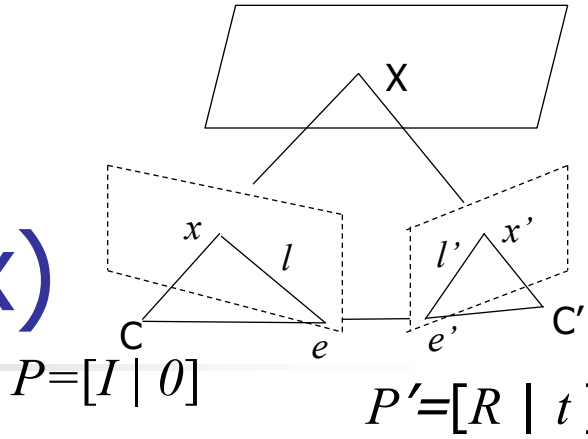
  Proof: For a skew symmetric matrix S, $X^{T}SX=0$, for all X.

  Now, $X^{T}P'^{T}FPX=(P'X)^{T}F(PX)=x'^{T}Fx=0$ (for any X in P^3, as F is the fundamental matrix). …
- F corresponds to $P=[I \mid 0]$ & $P'=[SF \mid e']$, where e' is the right epipole of F s.t. $e'^{T}F=0$.
- A good choice of $S=[e']_{x}$.
- F → $([I \mid 0], [[e']_{x}F \mid e'])$

  ←→ $([I \mid 0], [[e']_{x}F + e'\mathbf{v}^{T} \mid ke'])$

NPTEL

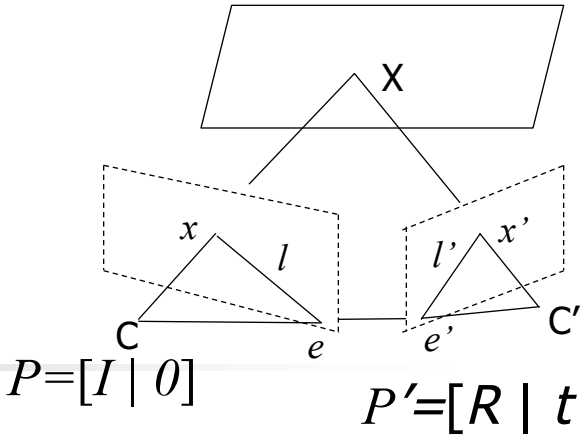
The camera matrices from E (Essential matrix)



- E is an essential matrix iff two of its singular values are equal and the third one is zero.
- $E=[t]_x R$
- $[t]_x$ and R can be computed through decomposition of E s.t. $E=SR$, where S is a skew symmetric matrix and R is orthogonal.

NPTEL

Decomposition of E (Essential matrix)



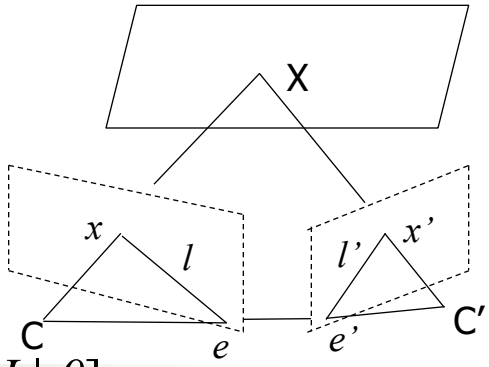
- SVD of $E=U$ diag(1,1,0) V^T
- Two possible decomposition of $E=SR$
- $S=UZU^T$ and $R=UWV^T$ or UW^TV^T

$$z = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad W = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Any skew symmetric matrix S can be decomposed as $S=kUZU^T$
- W is orthogonal and $Z=$diag(1,1,0)W.

NPTEL

Camera matrices from E



$P=[I \mid 0]$ $P'=[R \mid t]$

- SVD of $E=U$ diag(1,1,0) V^T
- Two possible decomposition of $E=SR$
- $S=UZU^T$ and $R=UWV^T$ or UW^TV^T

$$z=\begin{bmatrix}0 & 1 & 0\\ -1 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

$P'=[UWV^T \mid +u_3]$ or $[UWV^T \mid -u_3]$

$[UW^TV^T \mid +u_3]$ or $[UW^TV^T \mid -u_3]$

$$W=\begin{bmatrix}0 & -1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}$$

Last column of U.

Out of the four only one is valid for viewing a point from both the cameras. It is sufficient to test a single point for the above.

NPTEL

Ex. 5

- Check whether the following fundamental matrix and the camera matrices P and P′ (for left and right cameras) are compatible.

$$F = \begin{bmatrix} 20 & 12 & 0 \\ 8 & -7 & 59 \\ -4 & 33 & -177 \end{bmatrix}$$

$$P = \begin{bmatrix} 7 & 4 & -6 & 3 \\ 8 & -1 & 2 & -5 \\ 9 & -10 & 4 & 1 \end{bmatrix} \qquad P' = \begin{bmatrix} 6 & 4 & -6 & 10 \\ 8 & -5 & 2 & -7 \\ 9 & -10 & 6 & 2 \end{bmatrix}$$

NPTEL

Ans. 5

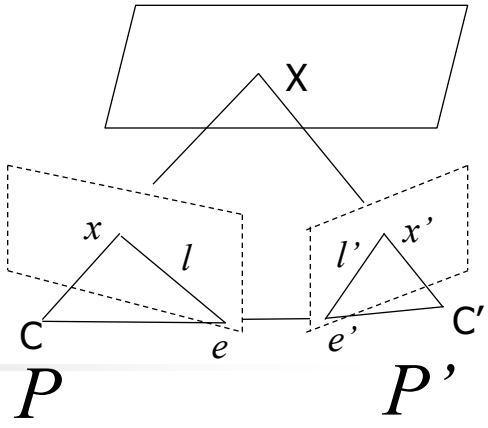
- $P'^{T}FP=$

$$\begin{bmatrix} -6549 & 11489 & -4746 & -2242 \\ 11859 & -14183 & 4926 & 2950 \\ -8496 & 8816 & -2784 & -1888 \\ -4071 & 7979 & -3414 & -1534 \end{bmatrix}$$

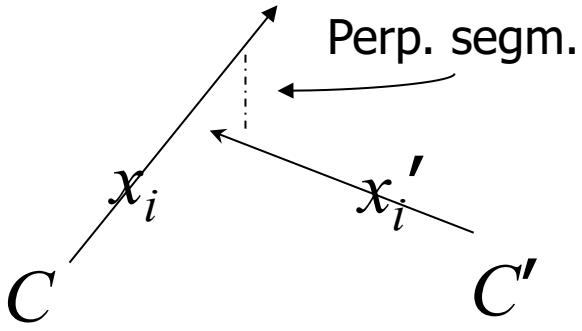
NPTEL

- Not a skew symmetric matrix.
- Not compatible.

Computing scene points (structure)



Given $x_i \leftrightarrow x_i'$, compute X.



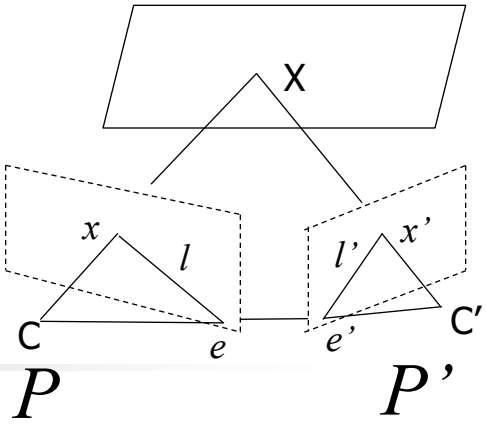
1. Compute F.
2. Compute P and P'.
3. For each (x_i, x_i') compute X by triangulation.

   i. Compute intersection of Cx_i and $C'x_i'$.
   ii. Compute segment perpendicular to both.
   iii. Get the mid-point.

Not projective invariant, i.e. $(PH, P'H)$ does not give $H^{-1}X$.

NPTEL

Minimizing Reprojection Error



Given $x_i \leftrightarrow x_i'$, compute X.

1. Estimate $\hat{X}$ s.t. $P\hat{X} = \hat{x}$ and $P'\hat{X} = \hat{x'}$.
2. Minimize the reprojection error (E_{rp}).

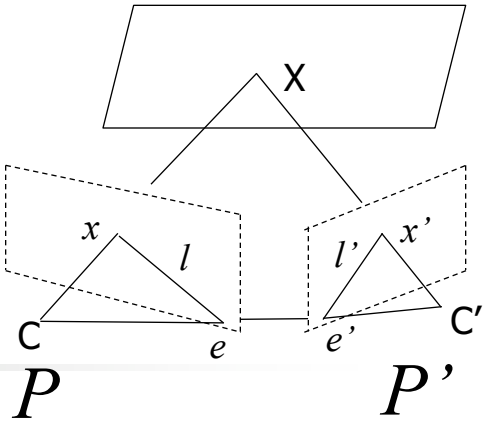
$$E_{rp} = d(x, \hat{x})^2 + d(x', \hat{x'})^2$$

subject to $x'^T F x = 0$

Projective invariant.

NPTEL

Linear triangulation methods



Given $x_i \leftrightarrow x_i'$, compute X.

$$x \times PX = 0$$
$$x' \times P'X = 0$$

4 equations,
3 unknowns.

$[A]_{4x4}X=0$

Minimize $||AX||$
subject to $||X||=1$.

Use DLT.
Not projective invariant.

Generalize to multiview correspondences.

$x_1 \leftrightarrow x_2 \leftrightarrow x_3$
$P_1 \quad P_2 \quad P_3$

$$x_1 \times P_1 X = 0$$
$$x_2 \times P_2 X = 0$$
$$x_3 \times P_3 X = 0$$

6 equations
3 unknowns.

NPTEL

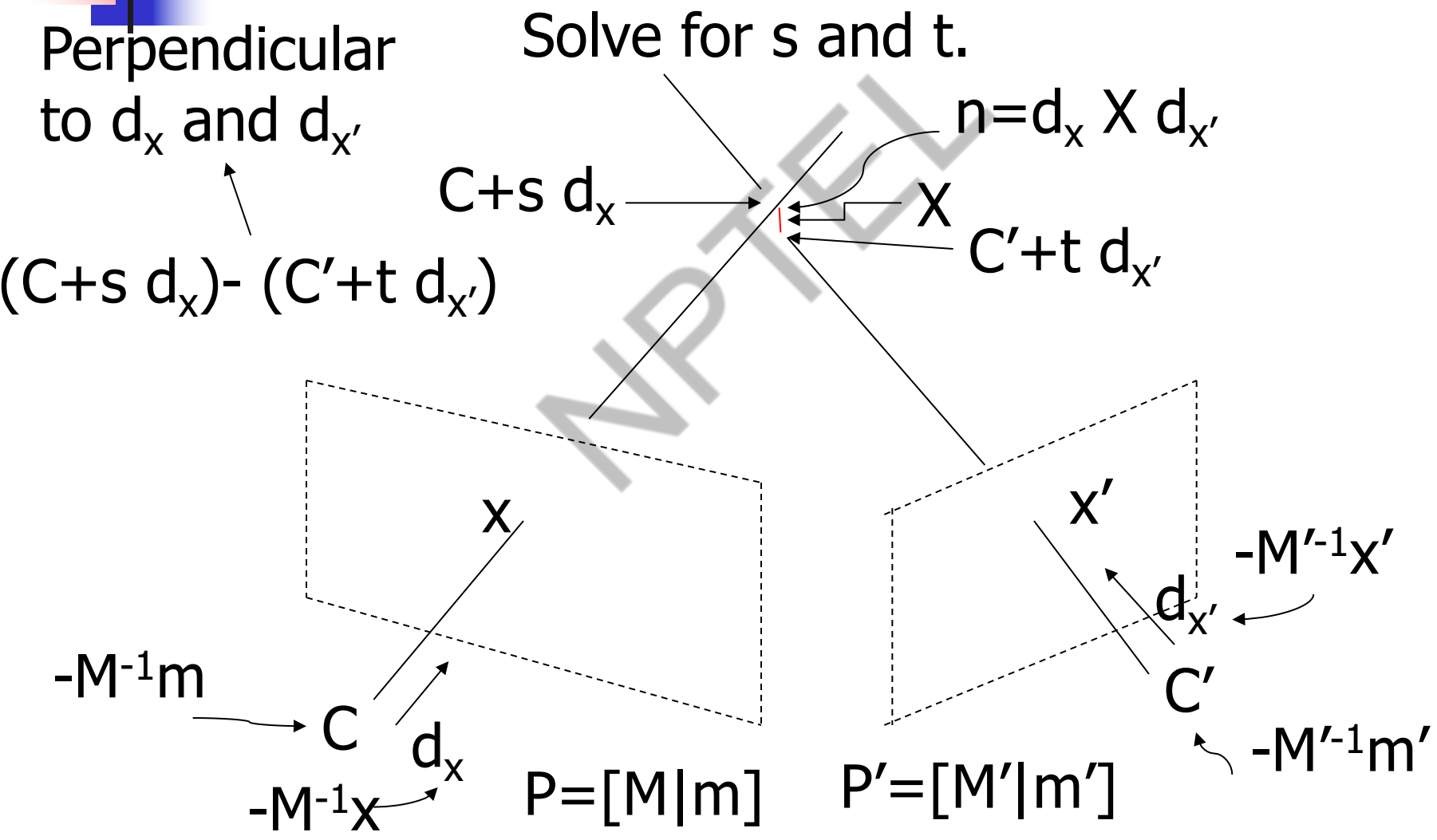
Ex. 4

- Suppose P and P’ (for left and right cameras). Images of a scene point are formed at (0,3.5), and (2/3, -1/3), respectively. Find the 3D coordinate of the scene point.

NPTEL

$$P = \begin{bmatrix} 7 & 4 & -6 & 3 \\ 8 & -1 & 2 & -5 \\ 9 & -10 & 4 & 1 \end{bmatrix} \quad P' = \begin{bmatrix} 6 & 4 & -6 & 10 \\ 8 & -5 & 2 & -7 \\ 9 & -10 & 6 & 2 \end{bmatrix}$$

Computation of structure



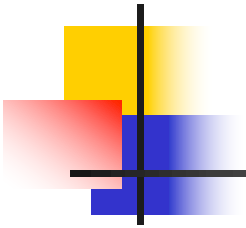
NPTEL

4 Ans.

$$P = \begin{bmatrix} 7 & 4 & -6 & 3 \\ 8 & -1 & 2 & -5 \\ 9 & -10 & 4 & 1 \end{bmatrix} \quad P' = \begin{bmatrix} 6 & 4 & -6 & 10 \\ 8 & -5 & 2 & -7 \\ 9 & -10 & 6 & 2 \end{bmatrix}$$

- P=[M|m] P′=[M′|m′]
- C=-M^{-1}m= $[0.35\ 1\ 1.62]^T$
- C′=-M'^{-1}m′=$[13\ 35\ \ 38]^T$
- x=$[0\ 3.5\ 1]^T$ x′=$[2/3\ -1/3\ 1]^T$
- d_x= M^{-1} x=[0.32 0.47 0.69]
- $d_{x'}$= M'^{-1} x′=[1.36 3.67 3.91]
- d_x X $d_{x'}$= [0.7 0.33 -0.55]

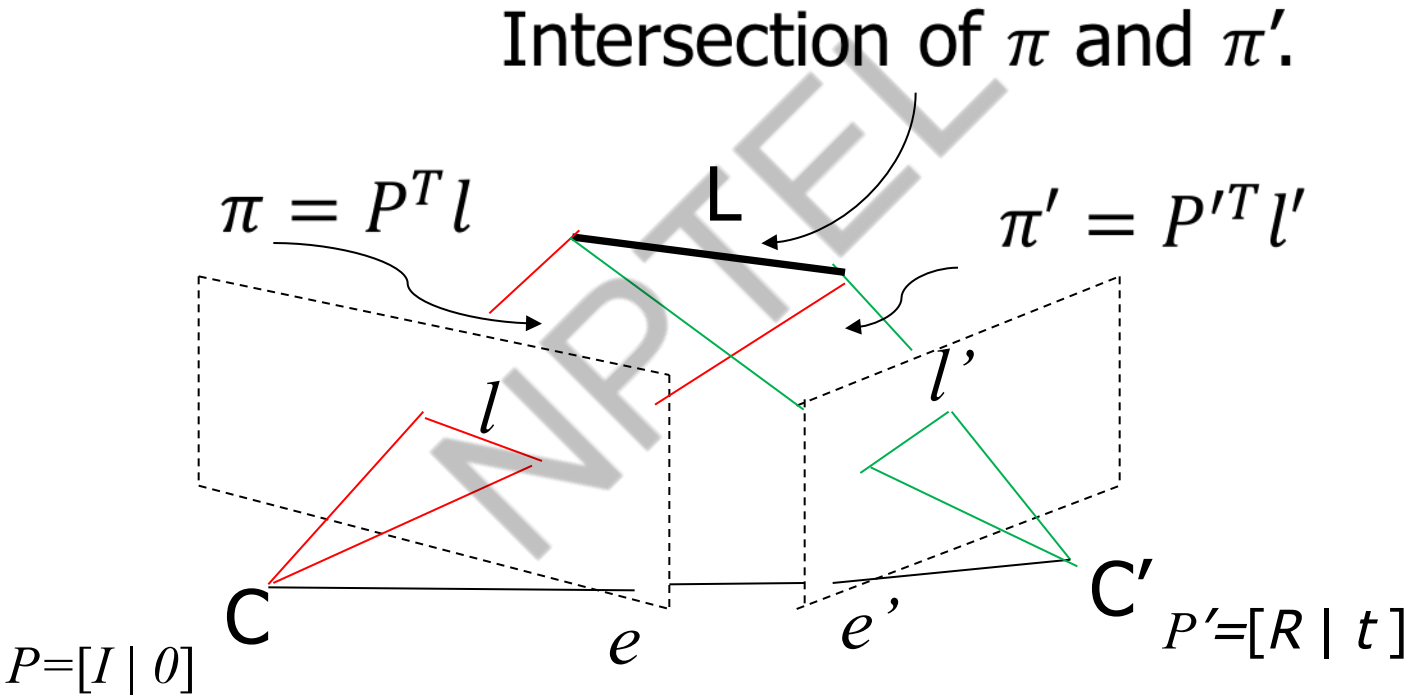
NPTEL



- $(C+s\ d_x)- (C'+t\ d_{x'})=[-12.85\ \ -33.93\ \ -36.58]+s[0.32\ \ 0.47\ \ \ 0.69]-t[1.36\ \ \ 3.67\ \ \ 3.91]$
- $0.32(-12.85+0.3s-1.36t)+0.47(-33.93+0.47s-3.67t)+0.69(-36.58+0.69s-3.91t)=0$ –(1)
- $1.36(-12.85+0.3s-1.36t)+3.67(-33.93+0.47s-3.67t)+3.91(-36.58+0.69s-3.91t)=0$ – (2)
- Solve (1) and (2) to get s and t, and the point.

NPTEL

Line reconstruction



$$L = \begin{bmatrix} \pi \\ \pi' \end{bmatrix}$$

A convenient way of representing 3D line.

NPTEL

Plane Induced Homography

$Proof$:

$$x' = P'X = [A|a]X$$

Now, $x = PX = [I\ |0]X$

So any point in $\overrightarrow{CX}$ is $X = \begin{bmatrix} x \\ \rho \end{bmatrix}$

When it intersects π, $\pi^T \begin{bmatrix} x \\ \rho \end{bmatrix} = 0$.

$$\Rightarrow \boldsymbol{v}^T x + \rho = 0$$

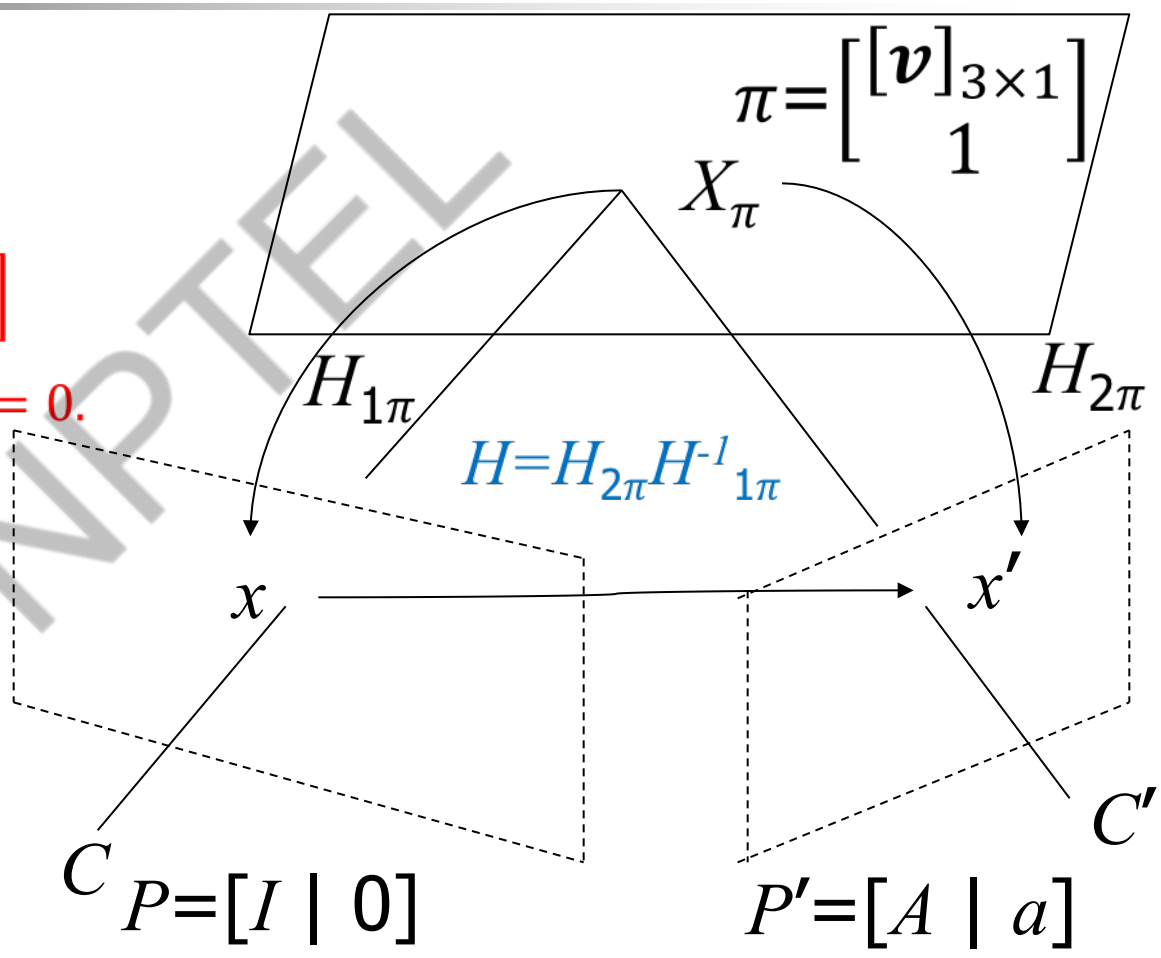
$$\Rightarrow \rho = -\boldsymbol{v}^T x$$

So, $x' = P'X = [A|a] \begin{bmatrix} x \\ -\boldsymbol{v}^T x \end{bmatrix}$

$= Ax - a\ \boldsymbol{v}^T x$

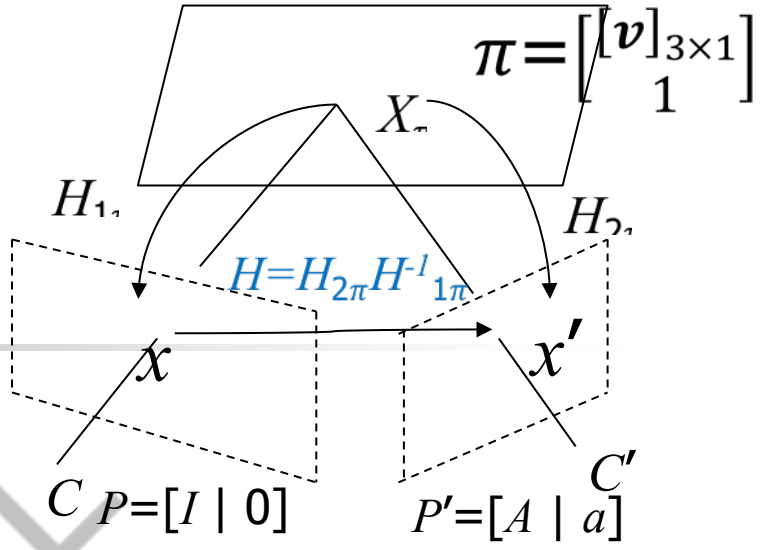
$=(A - a\ \boldsymbol{v}^T)x$

H



NPTEL

Plane induced homography



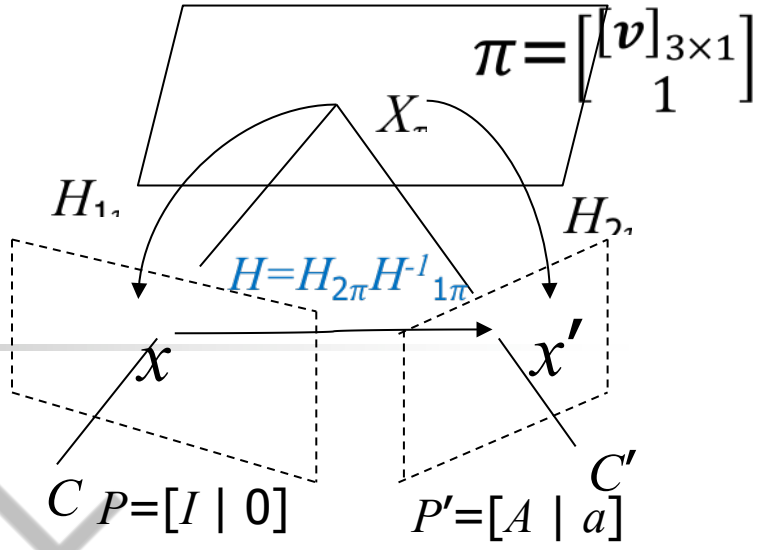
A transformation H between two stereo images is plane induced homography if F is decomposed into $[e']_x H$.

Hence, $P=[I|0]$ & $P'=[H|e']$.

Given $P=[I|0]$, $P'=[A|a]$, & a plane induced homography H, the plane can be recovered by solving $kH=A-av^T$, (linear equations for unknowns k and v).

NPTEL

Plane induced homography



A transformation H between two stereo images is plane induced homography if F is decomposed into $[e']_x H$.
Hence, $P=[I|0]$ & $P'=[H|e']$.

Plane at infinity

H is the transformation w.r.t. plane $[0\ 0\ 0\ 1]^T$ in the camera coordinate.

Given $P=[I|0]$, $P'=[A|a]$, & a plane induced homography H, the plane can be recovered by solving $kH=A-av^T$, (linear equations for unknowns k and v).

NPTEL

Homography compatible stereo geometry

H is compatible iff H^TF is skew symmetric, i.e.

$$H^TF+F^TH=0$$

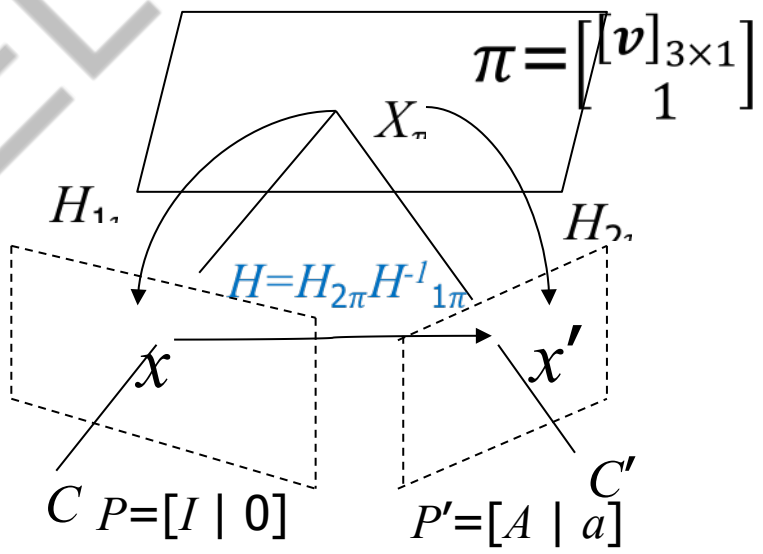
$$x'^T Fx = 0$$

And, $x' = Hx$

$$\Rightarrow (Hx)^T Fx = 0$$

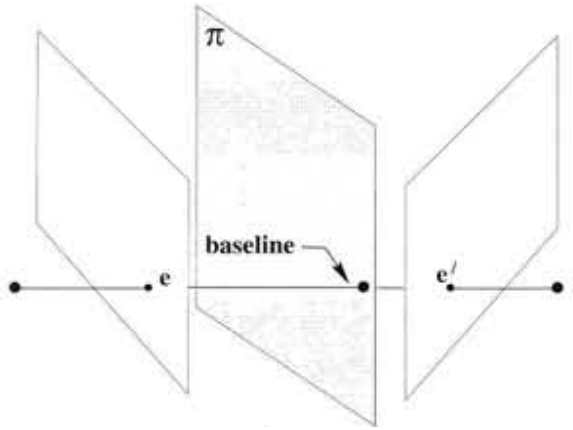
$$\Rightarrow x^T H^T Fx = 0$$

As this is true for all x, H^TF is skew symmetric.

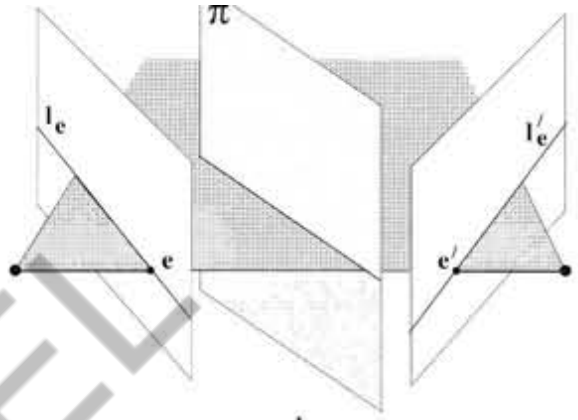


NPTEL

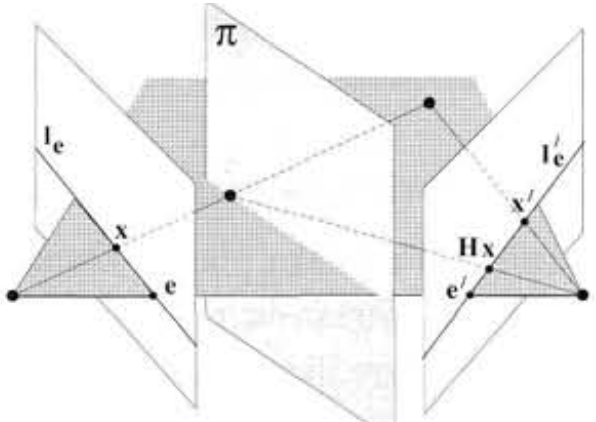
Plane induced H and epipolar constraints



Epipoles mapped by H, as e'=He, *since they are images of the point on the plane where the baseline intersects it.*



Epipolar lines are mapped by the homography as $H^{-T}l'_e = l_e$.

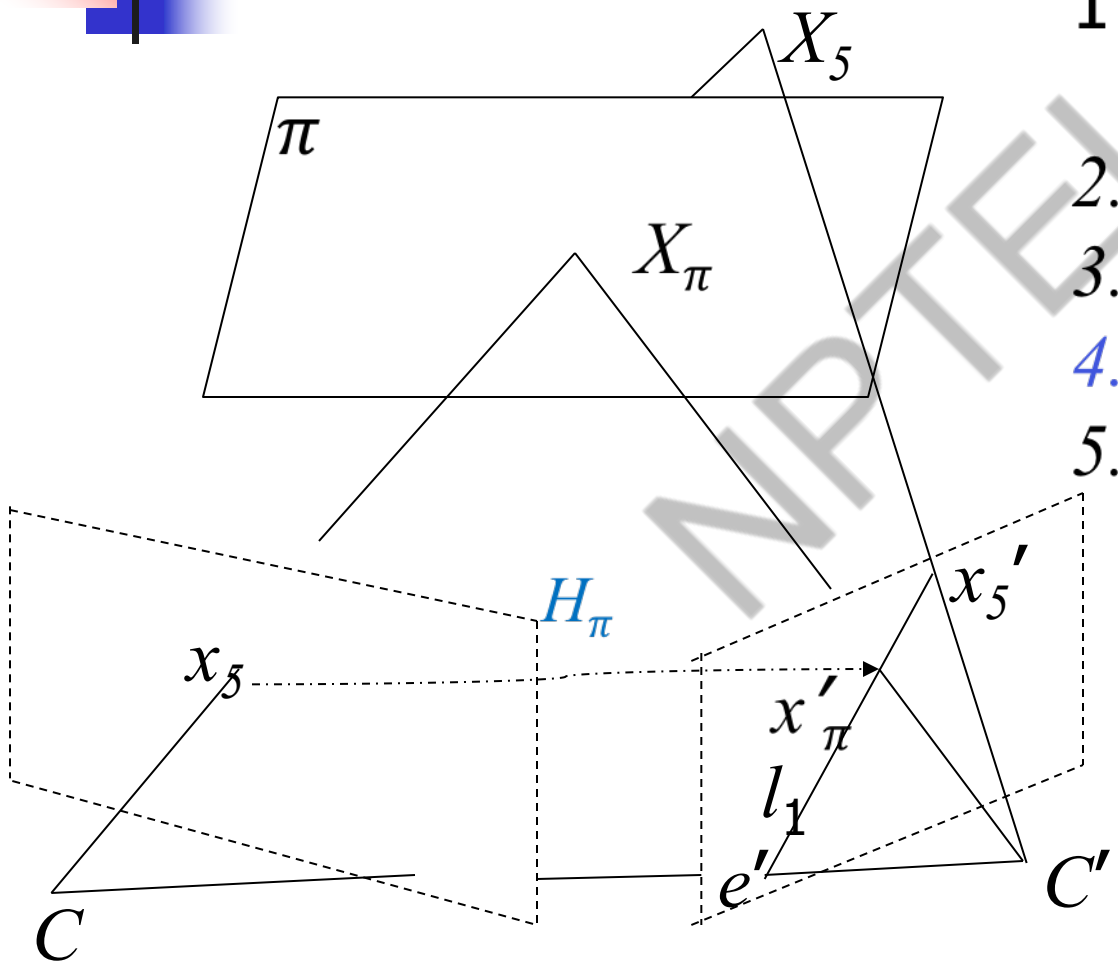


Hx *lies on epipolar line* l_e' *so* $l'_e = Fx = x' \times (Hx)$.

From Hartley and Zisserman, "Multiple view geometry in computer vision", Cambridge Univ. Press (2000)

NPTEL

Computing F from 6 points out of which 4 are coplanar



1. Use 4 coplanar points to compute H_π.
2. $l_1 = H_\pi(x_5) \times x_5'$
3. $l_2 = H_\pi(x_6) \times x_6'$
4. $e' = l_1 \times l_2$
5. $F = [e']_\times H_\pi$.

NPTEL

Given F and 3 point correspondences, compute H.

First Method

1. Obtain $(P=[I|0], P'=[A|a])$ from F and construct 3 scene points, $X_1, X_2, \& X_3$.
2. Obtain plane $(v^T, 1)^T$.
3. Compute $H=A-av^T$

Second Method

1. Obtain (e, e') from F.
2. Use 3 correspondences + (e, e'), to obtain H.

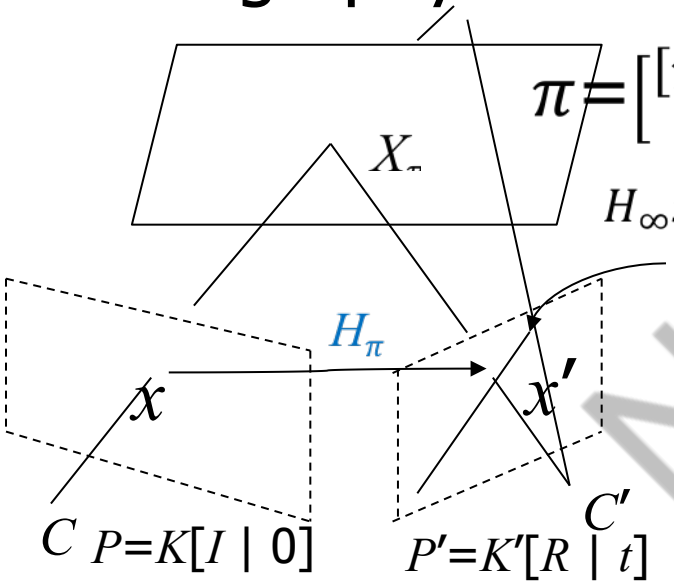
Any 3 points can bipartition the image space, w.r.t. the plane formed by them.

NPTEL

Infinite Homography

$H = (A - a\,\boldsymbol{v}^T)$
For $P=[I\,|0]$, $P'=[A \mid a]$
and plane$=[\boldsymbol{v}^T\ 1]^T$

Homography induced by the plane at α.



$$x' = K'RK^{-1}x + \frac{K't}{Z}$$

Vanishing point →

$$= H_\infty x + \frac{K't}{Z}$$

$$H_\pi = (K'R - e'\boldsymbol{v}^T)K^{-1}$$

where $v = \frac{n}{d}$ and $e' = K't$

$$H_\pi = K'\left(R - \frac{t\boldsymbol{n}^T}{d}\right)K^{-1}$$

$$= K'RK^{-1} - K'\frac{t\boldsymbol{n}^T}{d}K^{-1}$$

As $d \to \infty$, $H_\infty = K'RK^{-1}$

As $Z \to \infty$,
x' is the image of point on π_∞.

NPTEL

H_α and Vanishing points

- H_α maps vanishing points between two images.
- H_α can be computed by identifying three non-collinear vanishing points given F or from 4 vanishing points.
- Let $P=[M\,|m]$, $P'=[M'\,|\,m']$, $X=[x_\alpha{}^T\ 0]^T$ (a point at infinity).

$$x=PX=M\,x_\alpha$$
$$x'=P'X=M'\,x_\alpha$$
$$x'=M'M^{-1}x \rightarrow H_\alpha=M'M^{-1}$$

NPTEL

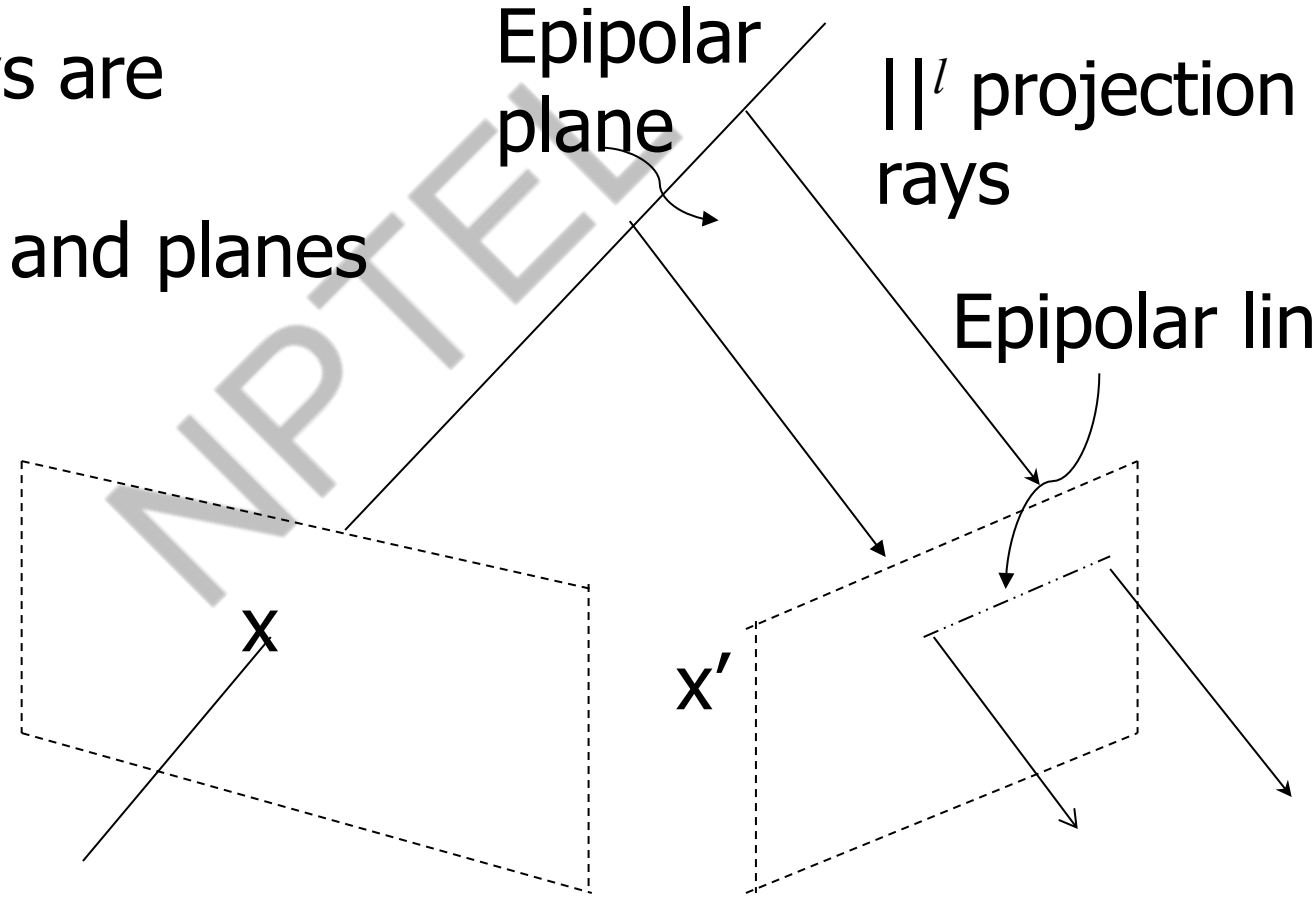
Affine epipolar geometry

- Projection rays are parallel.
- Epipolar lines and planes are parallel.

Form of F:

$$\begin{bmatrix} 0 & 0 & a \\ 0 & 0 & b \\ e & d & c \end{bmatrix}$$

a,b,c,d,e all non-zero.



As epipolar lines are parallel, epipoles are in the form $[e_1\ e_2\ 0]^T$

NPTEL

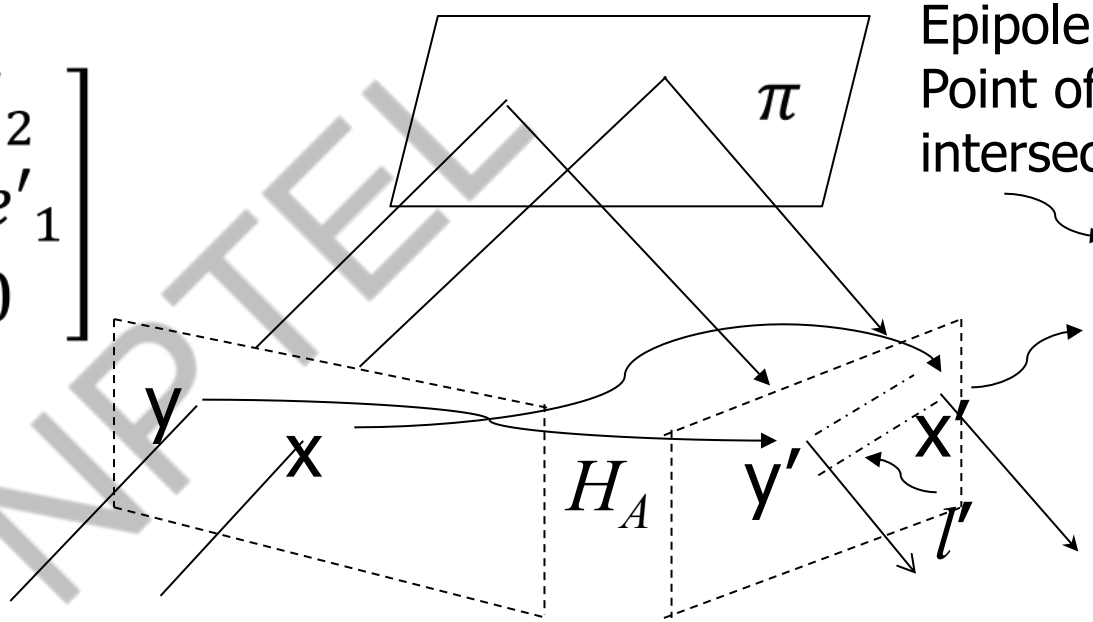
Affine stereo

$$\begin{bmatrix} 0 & 0 & a \\ 0 & 0 & b \\ e & d & c \end{bmatrix} \Rightarrow F_A = [e']_\times H_A$$

$$l' = e' \times H_A x = [e']_\times H_A x$$

$$[e']_\times = \begin{bmatrix} 0 & 0 & e'_2 \\ 0 & 0 & -e'_1 \\ -e'_2 & e'_1 & 0 \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \boldsymbol{b} \\ -\boldsymbol{b}^T & 0 \end{bmatrix}$$

$$H_A = \begin{bmatrix} A & \boldsymbol{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$$



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F_A: d.o.f.: 4

$$F_A = \begin{bmatrix} 0 & \boldsymbol{b} \\ -\boldsymbol{b}^T A & -\boldsymbol{b}^T \boldsymbol{t} \end{bmatrix} = \begin{bmatrix} 0 & 0 & e'_2 \\ 0 & 0 & -e'_1 \\ e & d & c \end{bmatrix}$$

Left epipole:$[-d\ e\ 0]^T$

right epipole:$[-b\ a\ 0]^T$

Estimating F_A

$$F_A = \begin{bmatrix} 0 & 0 & a \\ 0 & 0 & b \\ e & d & c \end{bmatrix}$$

Epipolar lines:

$$l' = F_A \boldsymbol{x} = [a \quad b \quad ex + dy + c]^T$$

$$l = {F_A}^T \boldsymbol{x}' = [e \quad d \quad ax' + by' + c]^T$$

Point correspondence: *Reduced to a single linear equation.*

$$x'^T F_A x = 0 \Longrightarrow ax' + by' + ex + dy + c = 0$$

$$[A]_{N\times 5} \boldsymbol{f}_{5\times 1} = 0$$

Solve using DLT.

Minimum 4 point correspondences required to get F_A.

Singularity constraint is satisfied by the structure of F_A.

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Estimating F_A (another approach)

$$F_A = \begin{bmatrix} 0 & 0 & a \\ 0 & 0 & b \\ e & d & c \end{bmatrix}$$

1. Compute H_A using 3 point-correspondences.
2. $l' = H_A x_4' \times x_4'$ (say, $[l_1\ l_2\ l_3]^T$)
3. Get e' from l' as $[l_2\ -l_1\ 0]$
4. $F_A = [e']_x H_A$

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Summary

- Epipolar geometry in a stereo imaging system.
  - epipoles, scene point, corresponding image points, and camera centers lie on the same plane.
- Fundamental matrix (F) characteristics and unique to the stereo set-up.
  - Transformation invariant.
  - 3x3 singular matrix with d.o.f. 7 and rank 2.
  - Given an image point x, its epipolar line l=Fx.
  - For any pair of correspondence point (x,x'), $x'^{T}Fx=0$.
  - Given epipoles (e,e'), $Fe=e'^{T}F=0$
  - Given camera matrices (P,P'), F is unique.
    - $P=[I|0]$, $P'=[M|m]$ ➔ $F=[m]_{x}M$.
    - $P=[M|m]$, $P'=[M'|m']$ ➔ $F=[m'-M'M^{-1}m]_{x}M'M^{-1}$

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Summary

- Given a homography H (4x4 non-singular matrix) in P^3, if $(\mathrm{P},\mathrm{P}') \rightarrow \mathrm{F}$, then $(\mathrm{PH},\mathrm{P}'\mathrm{H}) \rightarrow \mathrm{F}$.
- Given a fundamental matrix F, there exist a family of stereo setups (pairs of camera matrices).
  - $([I \mid 0],\ [[e']_{\mathrm{x}} F + e'v^T \mid ke'])$, v: any arbitrary 3 vector, k : a scalar constant.
- Given camera matrices (P,P′) and a corresponding pair of image points (x,x′), it is possible to reconstruct the respective 3-D scene point X.

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Summary

- Fundamental matrix of calibrated cameras is called Essential Matrix E.
- $E=[t]_x R$, given $([I|0], [R|t])$.
- E is an essential matrix iff two of its singular values are equal and the third one is zero.
- $[t]_x$ and R can be computed through decomposition of E s.t. $E=SR$, where S is a skew symmetric matrix and R is orthogonal.

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Summary

- SVD of $E=U$ diag(1,1,0) V^T
- Two possible decomposition of $E=SR$
- $S=UZU^T$ and $R=UWV^T$ or UW^TV^T

$$Z=\begin{bmatrix}0 & 1 & 0\\ -1 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

$P'=[UWV^T \mid +u_3]$ or $[UWV^T \mid -u_3]$

$[UW^TV^T \mid +u_3]$ or $[UW^TV^T \mid -u_3]$

$$W=\begin{bmatrix}0 & -1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}$$

One of the above is valid in viewing a point from both the cameras.

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Summary

- Given a set of pairs of corresponding points, it is possible to estimate F.
  - Minimum 7 pairs required.
- Parametric representation of F.

$$F = \begin{bmatrix} a & b & \alpha a + \beta b \\ c & d & \alpha c + \beta d \\ \alpha' a + \beta' c & \alpha' b + \beta' d & \alpha\alpha' a + \alpha\beta' c + \alpha'\beta b + \beta\beta' d \end{bmatrix}$$

$$Epipoles: \sout{e' = [\alpha \quad \beta \quad -1]^T} \quad e = [\alpha' \quad \beta' \quad -1]^T$$

$$e = [\alpha \quad \beta \quad -1]^T \quad e' = [\alpha' \quad \beta' \quad -1]^T$$

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Summary

- Given a set of pairs of corresponding points, it is possible to estimate camera matrices and scene points up to projective (4x4 homography matrix) ambiguity.
  - $(PH)(H^{-1}X) \leftrightarrow (P'H)(H^{-1}X)$
- Given a pair of corresponding lines l and l′, and camera matrices (P,P′) possible to reconstruct respective 3D line L.
  - Intersection of planes $P^T l$ and $P'^T l'$

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Summary

- A plane induces homography between corresponding image points in a stereo set-up.
  - Given a plane $(\mathbf{v}^T,1)$, and camera matrices $([\mathrm{I}|0],[\mathrm{A}|a])$, $H=(A-a\mathbf{v}^T)$
    - Homography at infinity: Plane at infinity $(\mathbf{0}^T,1)$ induces $H=A$.
- Affine epipolar geometry simplifies the structure of fundamental matrix.
  - Right epipole: $[-d\ e\ 0]^T$
  - Left epipole: $[-b\ a\ 0]^T$

$$F_A = \begin{bmatrix} 0 & 0 & a \\ 0 & 0 & b \\ e & d & c \end{bmatrix}$$

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